Power BI for Interviews

Complete Interview Training Guide

Level 1 (Basics) through Level 11 (Project Scenarios)

For Data Analyst, Power BI Analyst, Power BI Developer, Business Analyst, Product Analyst and MIS Analyst roles

Simple English answers with step-by-step Power BI techniques for every topic

ABOUT THIS GUIDE

Every answer in this guide is written the way a real person would say it in an interview: short, simple and clear. Read the Interview Answer, then say it again in your own words. Where an answer talks about a project, replace it with your own real project details. Interviewers can tell when an answer is memorized, so use your own experience.

SAMPLE TABLES USED IN EXAMPLES

Most examples use a simple sales model. You can build it in Power BI with a small Excel file and practice each step yourself.

Table	Columns
Sales (fact)	OrderID, OrderDate, ProductID, CustomerID, Qty, Price, Amount, Cost
Product (dimension)	ProductID, ProductName, Category
Customer (dimension)	CustomerID, Name, City, Region
Date (dimension)	Date, Year, Month No, Month, Quarter

HOW EACH TOPIC IS WRITTEN

Every topic has: Definition, Simple Explanation, Example (with DAX or Power Query code where useful), Technique (Step-by-Step clicks), Interview Answer, Follow-up Questions, Common Mistake and Related topics. Every level ends with a Quick Revision table, Top Interview Questions and Practical Scenarios.

WHICH LEVELS MATTER FOR YOUR ROLE

Role	Focus on	Be strong in
Junior / Fresher Data Analyst	Levels 1, 2, 3, 5, 7	Power Query cleaning, simple measures, clean charts
Power BI Analyst	Levels 3 to 8	Modeling, DAX, visuals, publishing and refresh
Power BI Developer	Levels 2 to 9	Star schema, DAX, RLS, refresh, performance
Business Analyst	Levels 7, 8, 10, 11	Requirements, KPIs, storytelling, basic DAX
Product Analyst	Levels 5, 6, 7, 10	Product metrics, funnel, retention, time intelligence
MIS Analyst	Levels 3, 7, 8, 10	Automated reports, refresh, subscriptions, accuracy

LEVEL 1 - POWER BI BASICS

What is Power BI?

Definition: Power BI is a Microsoft tool that turns raw data into reports and dashboards you can see and share.

Simple Explanation: Think of it like this: Excel holds your numbers, and Power BI helps you show them as clear charts that people can click and explore. It connects to many data sources, cleans the data, builds a model, and creates visuals.

Example: A manager has sales data in Excel. In Power BI you build one page that shows sales by month, by region and by product.

Technique (Step-by-Step): 1. Open Power BI Desktop. 2. Click Get data and pick your file. 3. Clean the data in Power Query. 4. Build charts on the Report page. 5. Click Publish to share it online.

Interview Answer: “Power BI is a tool I use to take data from places like Excel or a database, clean it, and show it as simple charts and dashboards. It helps managers see numbers quickly and take decisions.”

Follow-up Qs:
- **Q:** Is Power BI the same as Excel? **A:** No. Excel is mainly for calculations on a sheet. Power BI is built for large data, reports and sharing.
- **Q:** Who uses Power BI? **A:** Data analysts, business analysts, managers and MIS teams.

Common Mistake: Saying Power BI is only a charting tool. It also cleans data and builds a data model.

Related: Leads to Power BI components.

Power BI Components

Definition: Power BI has three main parts: Desktop, Service and Mobile.

Simple Explanation: Desktop is the free app on your computer where you build reports. Service (app.powerbi.com) is the website where you publish and share. Mobile is the phone app to view reports. A Gateway is a small program that lets the Service reach data stored inside your company network.

Example: You build a report in Desktop, publish it to the Service, and your manager opens it on the phone app.

Technique (Step-by-Step): 1. Build the report in Power BI Desktop. 2. Click Publish on the Home tab. 3. Choose a workspace. 4. Open app.powerbi.com to see the report online. 5. Install the mobile app and sign in to view it on your phone.

Interview Answer: “I build my reports in Power BI Desktop. Then I publish them to the Power BI Service so others can view them. People can also open them on the mobile app.”

Follow-up Qs:
- **Q:** Is Desktop free? **A:** Yes. Desktop is free to download. To share reports with others you usually need a Pro license.
- **Q:** What is a gateway? **A:** It is a bridge that lets the online Service refresh data from files or databases inside your company.

Common Mistake: Trying to build full reports in the Service. Desktop is the main place to build.

Related: Related to licenses and workflow.

Power BI Licenses

Definition: A license decides what you can do in Power BI: Free, Pro, Premium Per User (PPU) and Fabric/Premium capacity.

Simple Explanation: With a Free license you can build in Desktop and use My workspace for yourself. With Pro you can share and collaborate in workspaces. PPU gives Pro plus extra features. Capacity (Premium or Fabric) is for large companies and lets many viewers read without each needing Pro.

Example: Your team of 5 needs to share reports with each other. Everyone needs Pro.

Technique (Step-by-Step): 1. Check which license your company gave you in the Service (profile picture, then Account). 2. If you cannot share a report, check if you have Pro. 3. Ask your admin if you need an upgrade.

Interview Answer: “Desktop is free, but to share reports inside a workspace people normally need a Pro license. Bigger companies use Premium or Fabric capacity so viewers do not need individual licenses.”

Follow-up Qs:
- **Q:** Can a Free user view a shared report? **A:** Usually not, unless the report is in a Premium or Fabric capacity workspace.
- **Q:** What is PPU? **A:** Premium Per User. It gives Pro plus extra premium features for one user.

Common Mistake: Forgetting that both the person who shares and the person who views may need a license.

Related: Related to workspaces (Level 8).

Report View, Data View and Model View

Definition: These are the three views on the left side of Power BI Desktop.

Simple Explanation: Report view is where you build charts. Table (Data) view is where you look at the data in each table. Model view is where you see tables and connect them with relationships.

Example: You check the data in Table view, fix a relationship in Model view, and build a chart in Report view.

Technique (Step-by-Step): 1. Look at the left side of Desktop. 2. Click the first icon for Report view. 3. Click the second icon (table) for Data view. 4. Click the third icon for Model view.

Interview Answer: “I use Report view to build visuals, Data view to check my data, and Model view to connect the tables with relationships.”

Follow-up Qs:
- **Q:** Where do you create relationships? **A:** In Model view.
- **Q:** Where do you add a measure? **A:** In Report or Data view, with New measure on the Home tab.

Common Mistake: Building charts before checking the data in Table view.

Related: Leads to data modeling (Level 4).

Power BI Workflow

Definition: The normal steps to make a Power BI report from start to end.

Simple Explanation: Get the data, clean it, build the model, write measures, create visuals, publish, and refresh. This is what interviewers want to hear when they say 'explain your process'.

Example: A monthly sales report: get data from SQL, clean it, link tables, write Total Sales, build charts, publish, set daily refresh.

Technique (Step-by-Step): 1. Get data (Home, Get data). 2. Clean in Power Query (Transform data). 3. Connect tables in Model view. 4. Write measures with DAX. 5. Build visuals in Report view. 6. Publish to the Service. 7. Set up scheduled refresh.

Interview Answer: “First I understand what the business wants. Then I get the data, clean it in Power Query, build a simple model, write measures, create the visuals, and publish. After that I set up refresh so the report stays current.”

Follow-up Qs:
- **Q:** Which step takes the most time? **A:** Usually data cleaning and modeling.
- **Q:** What do you do before opening Power BI? **A:** I talk to the user to understand what questions the report must answer.

Common Mistake: Starting to build visuals before understanding the business question.

Related: Summary of the whole guide.

File Types and Semantic Model

Definition: A .pbix file is a full Power BI report with data. A .pbit is a template without data. A semantic model (earlier called dataset) is the data and model behind a report.

Simple Explanation: Many reports can use one semantic model. This keeps numbers the same across reports. Microsoft changed the name from dataset to semantic model, so you may hear both.

Example: One sales semantic model feeds the sales report, the finance report and the manager dashboard.

Technique (Step-by-Step): 1. Save your work with File, Save as to get a .pbix. 2. Use File, Export, Power BI template to get a .pbit. 3. After publishing, find the semantic model in the workspace next to the report.

Interview Answer: “A .pbix file holds the report and its data. When I publish, the model becomes a semantic model, which other reports can reuse so everyone sees the same numbers.”

Follow-up Qs:

- **Q:** What is the difference between pbix and pbit? **A:** pbix has data. pbit is a template with the structure but no data.
- **Q:** Is dataset the same as semantic model? **A:** Yes. It is the new name.

Common Mistake: Making a separate copy of the model for every report, so numbers differ.

Related: Leads to Live Connection (Level 2).

Level 1 - Quick Revision

Item	Simple meaning
Desktop	Free app to build reports
Service	Website to publish and share
Mobile	Phone app to view
Gateway	Bridge to on-premises data
Pro license	Needed to share in workspaces
Report / Data / Model view	Build / check data / connect tables
.pbix / .pbit	Report with data / template

Key rule: Build in Desktop, share in the Service.

Top Interview Questions: (1) What is Power BI and why use it? (2) Desktop vs Service. (3) Explain your Power BI workflow.

Practical Scenarios: (1) Manager wants a monthly sales report - get data, model, visuals, publish. (2) Team cannot open your report - check licenses and workspace access.

LEVEL 2 - GETTING DATA

Data Sources

Definition: A data source is where your data comes from, like Excel, CSV, SQL Server, a website or SharePoint.

Simple Explanation: Power BI can connect to hundreds of sources. In most jobs you will use Excel, CSV, SQL databases, SharePoint folders and sometimes web pages or APIs.

Example: Get monthly sales from an Excel file and customer data from a SQL database.

Technique (Step-by-Step): 1. Click Home, then Get data. 2. Choose the source (Excel, SQL Server, Folder, Web...). 3. Select the file or enter the server name. 4. In the Navigator, tick the tables you need. 5. Click Transform data to clean, or Load to use it directly.

Interview Answer: "I mostly connect to Excel, CSV and SQL databases. I pick the source from Get data, select the tables I need, and then clean them in Power Query before loading."

Follow-up Qs:
- **Q:** Can you use many sources in one report? **A:** Yes. You can combine Excel, SQL and web data in one model.
- **Q:** Transform data or Load? **A:** Transform opens Power Query to clean first. Load brings data in as it is.

Common Mistake: Loading data without cleaning it first.

Related: Leads to Import vs DirectQuery.

Import Mode

Definition: In Import mode, Power BI copies the data into the report file.

Simple Explanation: It is the default and the fastest mode, because the data lives inside Power BI. The downside is that data is only as new as the last refresh.

Example: A daily sales report that only needs fresh data once every morning.

Technique (Step-by-Step): 1. Get data and pick a source. 2. When asked, choose Import. 3. Load the data. 4. After publishing, set a scheduled refresh in the Service.

Interview Answer: "Import copies the data into Power BI, so the report is very fast. I use it most of the time, and set a scheduled refresh to keep the data up to date."

Follow-up Qs:
- **Q:** What is the main advantage? **A:** Speed, and all DAX features work.
- **Q:** What is the main limit? **A:** Data is old until the next refresh, and there are size limits.

Common Mistake: Using Import for a report that must show data every minute.

Related: Compare with DirectQuery.

DirectQuery

Definition: In DirectQuery, Power BI does not store the data. It asks the source every time someone opens or clicks the report.

Simple Explanation: Use it when data is very big or must be almost live. The downside is that reports can be slower, because every click sends a query to the database.

Example: A live orders dashboard on a big SQL database.

Technique (Step-by-Step): 1. Get data from a database like SQL Server. 2. Choose DirectQuery instead of Import. 3. Keep the model simple. 4. Make sure the database is fast and has indexes.

Interview Answer: "DirectQuery reads the data live from the database and does not store it. I use it only when data is too big to import or must be real-time, because it can make reports slower."

Follow-up Qs:
- **Q:** Why can DirectQuery be slow? **A:** Each click sends a new query to the source.
- **Q:** Does DirectQuery need refresh? **A:** No scheduled refresh for data, because it is always live.

Common Mistake: Choosing DirectQuery without a fast database behind it.

Related: Compare with Import.

Live Connection

Definition: Live Connection connects your report to an existing published semantic model (or Analysis Services) instead of loading data.

Simple Explanation: You can only build visuals and measures in the report. You cannot change the model. It is great for teams that share one central model.

Example: The BI team publishes one model. Analysts build their own reports on top of it.

Technique (Step-by-Step): 1. In Desktop, click Get data. 2. Choose Power BI semantic models. 3. Pick the published model. 4. Build visuals. Measures can be added in the report, but the tables stay locked.

Interview Answer: "With Live Connection I connect to a model someone already built, so everyone uses the same numbers. I only build visuals on top of it."

Follow-up Qs:
- **Q:** Can you edit the model in Live Connection? **A:** No. You can only add report-level measures and visuals.
- **Q:** Why use it? **A:** One model for many reports keeps numbers consistent.

Common Mistake: Trying to add new tables to a Live Connection report.

Related: Related to semantic models (Level 1).

Import vs DirectQuery vs Live Connection

Definition: A quick way to choose how to connect to data.

Simple Explanation: Import: data copied, fast, needs refresh. DirectQuery: data stays at source, live, slower. Live Connection: connect to an existing model, no modeling.

Example: Small Excel file: Import. Huge live database: DirectQuery. Company shared model: Live Connection.

Technique (Step-by-Step): 1. Ask: how big is the data? 2. Ask: how fresh must it be? 3. Ask: does a ready-made model already exist? 4. Choose: default to Import unless you have a clear reason.

Interview Answer: "I use Import by default because it is fast. I switch to DirectQuery when the data is huge or must be live. If a company model already exists, I use Live Connection."

Follow-up Qs:
- **Q:** What is the default choice? **A:** Import.
- **Q:** What is a Composite model? **A:** A model that uses both Import and DirectQuery tables together.

Common Mistake: Choosing DirectQuery just because it sounds more advanced.

Related: Summary of this level.

Level 2 - Quick Revision

Mode	Simple meaning
Import	Copy data in; fast; needs refresh
DirectQuery	Live from source; slower; no copy
Live Connection	Use an existing published model
Composite	Mix of Import and DirectQuery

Key rule: Use Import unless you have a clear reason not to.

Top Interview Questions: (1) Import vs DirectQuery. (2) What is Live Connection? (3) How do you decide which mode to use?

Practical Scenarios: (1) Small Excel report - Import. (2) Live stock or order data - DirectQuery. (3) Using the company's central model - Live Connection.

LEVEL 3 - POWER QUERY (DATA CLEANING)

Power Query Editor and Applied Steps

Definition: Power Query Editor is the window where you clean and shape data before loading it. Every action you take is saved as a step.

Simple Explanation: The steps appear on the right side under Applied Steps. When data refreshes, Power Query repeats all the steps in order. You can click any step to see the data at that point, or delete a step if it was wrong.

Example: You remove blank rows, fix data types and rename columns. Next month the new file goes through the same steps by itself.

Technique (Step-by-Step): 1. Click Home, then Transform data. 2. The Power Query Editor opens. 3. Use the ribbon to clean the data. 4. Watch the Applied Steps list on the right. 5. Click Close and Apply to load the data.

Interview Answer: “Power Query is where I clean my data. Every clean-up action is saved as a step, so when new data comes in, the same steps run again automatically. That saves me a lot of manual work.”

Follow-up Qs:
- **Q:** What are Applied Steps? **A:** A saved list of every change, in order.
- **Q:** Can you undo a step? **A:** Yes. Click the X next to the step.

Common Mistake: Changing data by hand in Excel every month instead of using Power Query steps.

Related: Base for all topics in this level.

Data Types

Definition: A data type tells Power BI what kind of value a column holds: text, whole number, decimal, date, true/false and so on.

Simple Explanation: Wrong data types are a common cause of wrong results. For example, if numbers are stored as text, you cannot add them.

Example: A Sales column is text, so Sum does not work. Change it to Decimal number.

Technique (Step-by-Step): 1. Click the small icon on the left of the column name. 2. Choose the right type, like Whole Number or Date. 3. Or select the column, then Transform tab, Data Type. 4. If asked to replace the current step, click Yes.

Interview Answer: “The first thing I check is data types. If a number is stored as text, my totals will not work. So I make sure dates are dates and numbers are numbers.”

Follow-up Qs:
- **Q:** What if a date shows as text? **A:** Change the type to Date, or use Using Locale if the date format is different.
- **Q:** Why do errors appear after changing type? **A:** Some values cannot be converted, like text in a number column.

Common Mistake: Not checking data types, then getting wrong totals.

Related: Leads to handling errors.

Remove Rows, Filter Rows and Remove Duplicates

Definition: These steps remove data you do not need.

Simple Explanation: Remove top rows deletes junk header lines. Remove blank rows deletes empty rows. Filter rows keeps only rows that match a condition. Remove duplicates keeps one copy of repeated rows.

Example: A file has 3 junk lines on top, blank rows, and repeated orders.

Technique (Step-by-Step): 1. Home tab, Remove Rows, then Remove Top Rows, or Remove Blank Rows. 2. Click the drop-down arrow on a column header to filter values. 3. Select the column that defines a duplicate. 4. Home tab, Remove Rows, Remove Duplicates.

Interview Answer: “I remove extra rows at the top, blank rows and duplicates in Power Query. Before removing duplicates I always choose the right column, like Order ID, so I do not delete real data.”

Follow-up Qs:
- **Q:** Which columns do you pick for duplicates? **A:** The ID column that should be unique, like Order ID.
- **Q:** Does it change the original file? **A:** No. Power Query never changes the source file.

Common Mistake: Removing duplicates using the wrong columns.

Related: Leads to cleaning text.

Split Column and Merge Columns

Definition: Split Column cuts one column into many. Merge Columns joins many columns into one.

Simple Explanation: Use split for things like Full Name into First and Last. Use merge to join City and State into one Address column.

Example: Split 'John Smith' into 'John' and 'Smith'.

Technique (Step-by-Step): 1. Select the column. 2. Transform tab, Split Column, By Delimiter (for example, space). 3. To merge, select columns with Ctrl, then Transform tab, Merge Columns. 4. Choose a separator and a new column name.

Interview Answer: “I use Split Column to break a full name into first and last names, and Merge Columns to join things like city and state. It is a quick way to shape the data.”

Follow-up Qs:
- **Q:** What delimiters can you use? **A:** Comma, space, dash or any custom character.
- **Q:** Can you split by number of characters? **A:** Yes, there is an option for that too.

Common Mistake: Forgetting to give merged columns a clear name.

Related: Similar to Excel Text to Columns.

Replace Values, Fill Down and Trim

Definition: These steps fix messy values.

Simple Explanation: Replace Values changes one value to another. Fill Down copies a value into the empty cells below it. Trim removes extra spaces. Clean removes hidden characters. Capitalize Each Word fixes names.

Example: A city column has 'delhi ' and 'Delhi'. Trim and Capitalize make them the same.

Technique (Step-by-Step): 1. Select the column. 2. Transform tab, Format, Trim (and Capitalize Each Word). 3. Transform tab, Replace Values to fix wrong spellings. 4. Transform tab, Fill, Down to fill blank cells under a heading.

Interview Answer: “I always trim extra spaces and fix the text case. Hidden spaces make the same city look like two different cities in charts.”

Follow-up Qs:
- **Q:** When do you use Fill Down? **A:** When a value shows only once at the top of a group and the cells below are empty.
- **Q:** Why trim text? **A:** Extra spaces make two equal values look different.

Common Mistake: Skipping Trim and getting duplicate categories in charts.

Related: Similar to Excel TRIM and Find & Replace.

Pivot and Unpivot Columns

Definition: Unpivot turns many columns into rows. Pivot turns rows into columns.

Simple Explanation: Unpivot is very important. If your file has a separate column for Jan, Feb and Mar, charts will not work well. Unpivot them into two columns: Month and Sales.

Example: Excel has columns Jan, Feb, Mar. Unpivot gives columns Month and Value.

Technique (Step-by-Step): 1. Select the columns to turn into rows (Jan, Feb, Mar). 2. Transform tab, Unpivot Columns. 3. Rename Attribute to Month and Value to Sales. 4. To go the other way, use Pivot Column.

Interview Answer: “If my Excel data has one column per month, I unpivot it so I get one Month column and one Value column. That makes it easy to build charts.”

Follow-up Qs:
- **Q:** Unpivot Columns vs Unpivot Other Columns? **A:** Unpivot Other Columns keeps the selected columns fixed and unpivots all others, which is safer when new months are added.
- **Q:** When do you use Pivot? **A:** When you need values from rows to become column headings.

Common Mistake: Building charts on wide data with one column per month.

Related: Important for modeling.

Group By

Definition: Group By summarizes rows into groups, like total sales per customer.

Simple Explanation: It works like a pivot table inside Power Query. You choose the column to group by and the calculation, such as Sum or Count.

Example: Turn a long order list into total sales per customer.

Technique (Step-by-Step): 1. Transform tab, Group By. 2. Choose the column to group on (Customer). 3. Give the new column a name (Total Sales). 4. Choose the operation (Sum) and the column to sum. 5. Click OK.

Interview Answer: “Group By in Power Query summarizes data, like total sales per customer, before it loads. I use it when I do not need every single row in the model.”

Follow-up Qs:
- **Q:** Group By vs DAX measure? **A:** Group By changes the data itself. A measure calculates on the fly and responds to filters. I prefer measures for most reports.
- **Q:** Can you group by many columns? **A:** Yes, with the Advanced option.

Common Mistake: Grouping the data too early and losing details you need later.

Related: Compare with DAX measures (Level 5).

Conditional Column and Custom Column

Definition: A conditional column adds a new column using if-then rules. A custom column uses a formula.

Simple Explanation: Conditional column is click-based and easy. Custom column uses M language for more control.

Example: Add a column 'Size': High if Sales is above 1000, else Low.

```
if [Sales] > 1000 then "High" else "Low"
```

Technique (Step-by-Step): 1. Add Column tab, Conditional Column. 2. Write the rule: If Sales is greater than 1000 then High. 3. Add Else value Low. 4. Click OK. 5. For a formula, choose Custom Column and type it.

Interview Answer: “I use Conditional Column to create groups, like High and Low sales, without writing code. For special cases I write a small formula in Custom Column.”

Follow-up Qs:
- **Q:** Conditional column vs DAX calculated column? **A:** Power Query columns are made before loading and are better for speed. DAX columns are made after loading.
- **Q:** Is M language case-sensitive? **A:** Yes.

Common Mistake: Writing M formulas with wrong capital letters.

Related: Related to calculated columns (Level 5).

Merge Queries vs Append Queries

Definition: Merge joins two tables side by side using a matching column. Append stacks two tables on top of each other.

Simple Explanation: Merge is like VLOOKUP or a SQL join. Append is for combining the same type of data, like sales from Jan, Feb and Mar files.

Example: Merge Orders with Customers to get the customer city. Append 12 monthly sales tables into one.

Technique (Step-by-Step): 1. For merge: Home tab, Merge Queries. 2. Pick both tables and click the matching column in each. 3. Choose the join kind (Left Outer is common). 4. Click OK and expand the new column. 5. For append: Home tab, Append Queries, pick the tables.

Interview Answer: “I use Merge when I need extra columns from another table, like a lookup. I use Append when I need to put the same kind of tables together, like monthly files.”

Follow-up Qs:
- **Q:** Which join kind do you use most? **A:** Left Outer, because it keeps all rows from the first table.
- **Q:** Merge vs Append? **A:** Merge adds columns. Append adds rows.

Common Mistake: Using Append when Merge was needed.

Related: Similar to SQL JOIN and UNION.

Reference vs Duplicate

Definition: Both make a copy of a query. Reference stays linked to the original. Duplicate is a full independent copy.

Simple Explanation: With Reference, if you change the original, the copy updates too. With Duplicate, the copy has all the same steps but lives on its own.

Example: Create a clean Sales query, then Reference it to make a summary query.

Technique (Step-by-Step): 1. Right-click the query in the left pane. 2. Choose Reference to make a linked copy. 3. Choose Duplicate to make an independent copy. 4. Rename the new query clearly.

Interview Answer: “I use Reference when I want to build on a cleaned query without repeating the steps. Duplicate is for when I want a separate version that will change on its own.”

Follow-up Qs:
- **Q:** When to use Reference? **A:** When the new query depends on the same cleaned base data.
- **Q:** What if you change the original query? **A:** A Reference copy changes too. A Duplicate does not.

Common Mistake: Using Duplicate and then forgetting to apply the same fix in both queries.

Related: Related to Merge and Append.

Combine Files from a Folder

Definition: This loads many files with the same layout (like monthly sales files) into one table.

Simple Explanation: You put all files in one folder. Power Query reads them all. When a new file is added to the folder, refresh brings it in.

Example: A folder with Jan.csv, Feb.csv and Mar.csv. All become one Sales table.

Technique (Step-by-Step): 1. Get data, Folder. 2. Enter the folder path. 3. Click Combine, then Combine and Transform Data. 4. Choose a sample file. 5. Clean the data and load. 6. Next month, drop in the new file and refresh.

Interview Answer: “Instead of opening each monthly file, I use Get data from Folder. Power Query combines them all. Next month I just add the new file to the folder and click Refresh.”

Follow-up Qs:
- **Q:** Do files need the same layout? **A:** Yes. Columns should be the same.
- **Q:** How are new files picked up? **A:** On refresh, Power Query reads the folder again.

Common Mistake: Putting a file with a different layout in the folder.

Related: Related to Append.

Handling Errors and Null Values

Definition: Errors are cells Power Query cannot calculate. Nulls are empty values.

Simple Explanation: You can remove errors, replace them with a value, or fix the cause. For nulls you can replace them with 0 or a label like 'Unknown', or fill them from other cells.

Example: A Price column has text 'N/A' that becomes an error when changed to number.

Technique (Step-by-Step): 1. Right-click the column, choose Remove Errors or Replace Errors. 2. For nulls, right-click, Replace Values, and replace null with 0 or Unknown. 3. Check the source to find out why errors came up. 4. Look at the column quality bar at the top.

Interview Answer: “I check each column's quality bar for errors and empty values. I fix the cause if I can. Otherwise I replace nulls with 0 or Unknown, and I tell the business what I did.”

Follow-up Qs:
- **Q:** How do you find errors quickly? **A:** Turn on Column quality and Column profile in the View tab.
- **Q:** Is replacing nulls with 0 always right? **A:** No. Sometimes null means unknown, and 0 would be wrong. Ask the business.

Common Mistake: Replacing nulls with 0 without checking what it means.

Related: Related to data types.

Level 3 - Quick Revision

Step	What it does
Applied Steps	Saved list of every clean-up action
Data type	Tells Power BI what kind of value a column has
Remove Duplicates	Keeps one copy of repeated rows
Split / Merge Columns	Cut one column / join columns
Unpivot	Turns many columns into rows
Group By	Summarize rows into groups
Merge	Join tables side by side
Append	Stack tables on top
Reference	Linked copy of a query
Duplicate	Independent copy of a query

Key rule: Clean data in Power Query. Calculate in DAX.

Top Interview Questions: (1) Merge vs Append. (2) Reference vs Duplicate. (3) What is Unpivot and why use it? (4) How do you clean messy data?

Practical Scenarios: (1) Excel has one column per month - Unpivot. (2) 12 monthly files in a folder - Combine from Folder. (3) Add customer city to orders - Merge Queries.

LEVEL 4 - DATA MODELING

Fact Table and Dimension Table

Definition: A fact table holds the numbers you measure, like sales. A dimension table holds the details that describe them, like customer, product and date.

Simple Explanation: Fact tables are long, with many rows. Dimension tables are short, with one row per item. Keeping them separate makes the model clean and fast.

Example: Sales (fact): OrderID, Date, ProductID, CustomerID, Amount. Product (dimension): ProductID, Name, Category.

Technique (Step-by-Step): 1. Find the table with numbers and events (orders, sales). That is the fact table. 2. Find tables that describe things (products, customers, dates). Those are dimensions. 3. Make sure each dimension has one unique key column.

Interview Answer: “A fact table stores the numbers, like sales amount and quantity. Dimension tables store the descriptions, like product name or customer city. I keep them separate so the model stays simple and fast.”

Follow-up Qs:
- **Q:** Which table is bigger? **A:** The fact table, usually.
- **Q:** Give examples of dimensions. **A:** Date, Product, Customer, Region.

Common Mistake: Putting everything into one big flat table.

Related: Leads to star schema.

Star Schema

Definition: A star schema has one fact table in the middle, connected to dimension tables around it, like a star.

Simple Explanation: It is the best-practice design for Power BI. It is simple, fast and easy to understand. A snowflake schema is when dimensions are split into more tables, which makes it more complicated.

Example: Sales in the middle, connected to Product, Customer, Date and Region.

Technique (Step-by-Step): 1. Put the fact table in the middle of Model view. 2. Place dimension tables around it. 3. Connect each dimension to the fact table with a one-to-many relationship. 4. Avoid linking dimensions to each other.

Interview Answer: “I design my model as a star schema, with a sales fact table in the center and dimension tables around it. It makes DAX easier and the reports faster.”

Follow-up Qs:
- **Q:** Star vs snowflake? **A:** Star keeps each dimension in one table. Snowflake splits dimensions into several tables.
- **Q:** Why is star schema recommended? **A:** It is simple, fast and gives fewer mistakes in DAX.

Common Mistake: Creating a messy web of relationships between many tables.

Related: Related to relationships.

Relationships and Cardinality

Definition: A relationship connects two tables using a matching column. Cardinality says how rows match: one-to-many, many-to-one, one-to-one or many-to-many.

Simple Explanation: One-to-many is the most common: one product can appear in many sales rows. The 'one' side must have unique values.

Example: Product[ProductID] (one) to Sales[ProductID] (many).

Technique (Step-by-Step): 1. Open Model view. 2. Drag the key column from one table to the matching column in the other. 3. Double-click the line to check the cardinality. 4. Make sure the one side has unique values.

Interview Answer: “I connect tables with relationships using key columns, like ProductID. Most of mine are one-to-many, from the dimension table to the fact table.”

Follow-up Qs:
- **Q:** What if the key is not unique on the one side? **A:** Power BI cannot make a proper one-to-many relationship. Clean the data first.
- **Q:** Why avoid many-to-many? **A:** It can give confusing numbers. A bridge table is usually better.

Common Mistake: Linking on a column that has duplicates on the 'one' side.

Related: Leads to cross-filter direction.

Cross-Filter Direction

Definition: Cross-filter direction decides which way filters flow between tables: Single or Both.

Simple Explanation: Single means the filter flows from the dimension to the fact table, which is the normal way. Both means filters flow in both directions. It can be useful, but it can also slow down the report and cause strange results.

Example: A Product slicer filters Sales. That is single direction.

Technique (Step-by-Step): 1. Double-click the relationship line in Model view. 2. Look at Cross-filter direction. 3. Keep it on Single in most cases. 4. Use Both only when you really need it, and test the results.

Interview Answer: “I keep the cross-filter direction as Single most of the time. I only use Both when there is a clear need, because it can slow things down and create unexpected results.”

Follow-up Qs:
- **Q:** Which is the default? **A:** Single, from the one side to the many side.
- **Q:** Why be careful with Both? **A:** It can create loops and slow performance.

Common Mistake: Setting everything to Both direction to 'make it work'.

Related: Related to active vs inactive relationships.

Active vs Inactive Relationships

Definition: Only one relationship between two tables can be active. The others are inactive, shown with a dotted line.

Simple Explanation: A good example: a Sales table has Order Date and Ship Date, both linked to the Date table. One is active. The other can be used in a measure when you need it.

Example: Active: Order Date to Date. Inactive: Ship Date to Date.

```
Sales by Ship Date =
CALCULATE([Total Sales],
    USERELATIONSHIP(Sales[ShipDate], 'Date'[Date]))
```

Technique (Step-by-Step): 1. Create both relationships in Model view. 2. Power BI keeps one active and makes the other dotted. 3. Write a measure with USERELATIONSHIP to use the inactive one when needed.

Interview Answer: “When one table has two date columns, like order date and ship date, only one relationship can be active. I use USERELATIONSHIP in a measure to use the other one.”

Follow-up Qs:
- **Q:** How many active relationships between two tables? **A:** Only one.
- **Q:** How do you use an inactive one? **A:** With USERELATIONSHIP inside CALCULATE.

Common Mistake: Making a second relationship active by mistake and breaking the first one.

Related: Related to date table.

Date Table

Definition: A date table is a separate table with one row for every date. It is needed for time calculations like year to date and last year.

Simple Explanation: You build one table with a continuous list of dates, add columns like Year, Month and Quarter, connect it to your fact table, and mark it as a Date table.

Example: A Date table from 2022 to 2025 with Year, Month Name, Month Number and Quarter.

```
Date = CALENDAR(DATE(2022,1,1), DATE(2025,12,31))

Year = YEAR('Date'[Date])
Month No = MONTH('Date'[Date])
Month = FORMAT('Date'[Date], "MMM")
```

Technique (Step-by-Step): 1. Modeling tab, New table. 2. Type the CALENDAR formula. 3. Add columns for Year, Month Number and Month Name. 4. Connect Date[Date] to Sales[OrderDate]. 5. Right-click the table, Mark as date table.

Interview Answer: “I create a proper date table with all dates and link it to my fact table. Without it, time functions like year-to-date and last year do not work correctly.”

Follow-up Qs:
- **Q:** CALENDAR vs CALENDARAUTO? **A:** CALENDAR needs start and end dates. CALENDARAUTO finds dates from your data.
- **Q:** Why not use the date column in the fact table? **A:** It may have gaps and repeats, so time functions can give wrong results.

Common Mistake: Using the date column from the sales table instead of a proper date table.

Related: Needed for time intelligence (Level 6).

Hierarchies

Definition: A hierarchy is a ready-made path of fields, like Year, Quarter, Month, Day or Country, State, City.

Simple Explanation: It lets users drill down in a chart with one click, from a big level to a smaller one.

Example: A Date hierarchy: Year to Quarter to Month.

Technique (Step-by-Step): 1. In the Data pane, right-click a field, choose Create hierarchy. 2. Drag other fields (Quarter, Month) under it. 3. Use the hierarchy in a visual. 4. Use the drill arrows on the visual to go down.

Interview Answer: “I create hierarchies like Year, Quarter and Month so users can drill down in a chart with a click.”

Follow-up Qs:
- **Q:** What is a hierarchy used for? **A:** Drill down and drill up.
- **Q:** Do dates already have a hierarchy? **A:** Yes, Power BI can auto-create one, but a proper Date table is better.

Common Mistake: Making hierarchies from unrelated fields.

Related: Related to drill down (Level 7).

Sort by Column

Definition: Sort by Column lets you sort one column using another column. For example, month names sort A to Z by default, which is wrong.

Simple Explanation: You sort the Month Name column by the Month Number column, so you get Jan, Feb, Mar and not Apr, Aug, Dec.

Example: Without it: Apr, Aug, Dec. With it: Jan, Feb, Mar.

Technique (Step-by-Step): 1. Click the Month Name column in Data view. 2. Column tools tab, Sort by column. 3. Choose Month No. 4. Check your chart. The months are now in the right order.

Interview Answer: “Month names sort alphabetically by default, so I use Sort by Column and sort them by month number to get January to December.”

Follow-up Qs:
- **Q:** Why do months look in the wrong order? **A:** They are text, so they sort alphabetically.
- **Q:** What do you need for it? **A:** A number column that gives the right order.

Common Mistake: Forgetting to set this and showing months in alphabetical order in charts.

Related: Related to date table.

Level 4 - Quick Revision

Concept	Simple meaning
Fact table	Numbers and events (sales)
Dimension table	Details (product, customer, date)
Star schema	Fact in the middle, dimensions around
One-to-many	Most common relationship
Single direction	Filter flows from dimension to fact
Inactive relationship	Dotted line; use USERELATIONSHIP
Date table	One row per date; needed for time functions
Sort by column	Fix month order

Key rule: Use a star schema and a proper date table.

Top Interview Questions: (1) What is a star schema? (2) Fact vs dimension tables. (3) What is a date table and why needed? (4) What is cardinality?

Practical Scenarios: (1) Sales has order date and ship date - two relationships, USERELATIONSHIP. (2) Months show in wrong order - Sort by Column. (3) Model is slow and messy - rebuild as a star schema.

LEVEL 5 - DAX BASICS

Calculated Column vs Measure vs Calculated Table

Definition: A calculated column adds a new column to a table. A measure calculates a value that changes with filters. A calculated table creates a new table with DAX.

Simple Explanation: A column is calculated row by row when data loads and is stored in the model. A measure is calculated when you use it in a chart, so it reacts to slicers and filters. In most cases, use a measure.

Example: Profit per row: calculated column. Total Profit shown in a card: measure.

```
Profit = Sales[Amount] - Sales[Cost]        -- calculated column

Total Sales = SUM(Sales[Amount])             -- measure
```

Technique (Step-by-Step): 1. For a measure: Home tab, New measure, type the formula. 2. For a column: Table tools, New column. 3. For a table: Modeling tab, New table. 4. Ask yourself: does the result need to change when a user clicks a slicer? If yes, use a measure.

Interview Answer: “A calculated column is worked out row by row and saved in the model. A measure is worked out when it is used in a visual, so it changes with filters. I use measures whenever possible, because they are lighter and more flexible.”

Follow-up Qs:
- **Q:** Which uses more memory? **A:** A calculated column, because it is stored.
- **Q:** Can a measure be used as a slicer? **A:** No. A slicer needs a column.

Common Mistake: Using a calculated column for totals, which will not react to filters.

Related: Leads to row and filter context.

Row Context and Filter Context

Definition: Row context means 'work on this one row'. Filter context means 'work on the rows left after filters'.

Simple Explanation: In a calculated column, DAX looks at one row at a time (row context). In a measure inside a chart, DAX looks at the rows left after slicers, filters and chart axes (filter context). This is the heart of DAX.

Example: In a bar chart by Region, the measure Total Sales gets a different filter context for each bar: one region at a time.

Technique (Step-by-Step): 1. Remember: a column works row by row. 2. A measure works on whatever rows are visible after filters. 3. Test a measure in a card, then in a chart by Region, and see how the number changes for each bar.

Interview Answer: “Filter context is the set of filters on the data at the moment of calculation, like a slicer or a chart axis. That is why one measure shows different results for each bar of a chart.”

Follow-up Qs:
- **Q:** Which context does a measure use? **A:** Filter context.
- **Q:** Which context does a calculated column use? **A:** Row context.

Common Mistake: Not understanding why the same measure gives different numbers in different visuals.

Related: Leads to CALCULATE.

Basic Aggregations (SUM, AVERAGE, COUNT, DISTINCTCOUNT, MIN, MAX)

Definition: These functions give one number from a column.

Simple Explanation: SUM adds numbers. AVERAGE finds the mean. COUNT counts numbers. COUNTROWS counts rows. DISTINCTCOUNT counts unique values. MIN and MAX give the smallest and largest value.

Example: Total sales, average order value, number of customers.

```
Total Sales = SUM(Sales[Amount])
Avg Order = AVERAGE(Sales[Amount])
Orders = COUNTROWS(Sales)
Customers = DISTINCTCOUNT(Sales[CustomerID])
```

Technique (Step-by-Step): 1. Click New measure. 2. Type the name, an equal sign and the function. 3. Pick the column. 4. Drop the measure in a Card visual to check the result.

Interview Answer: “These are the starting point of every report: SUM for totals, COUNTROWS for orders, DISTINCTCOUNT for unique customers. I always put them in measures.”

Follow-up Qs:
- **Q:** COUNT vs COUNTROWS vs DISTINCTCOUNT? **A:** COUNT counts values in a column. COUNTROWS counts rows in a table. DISTINCTCOUNT counts unique values.
- **Q:** Do they ignore blanks? **A:** COUNT and DISTINCTCOUNT skip blanks. COUNTROWS counts all rows.

Common Mistake: Using SUM on a column that has the wrong data type.

Related: Leads to IF and SWITCH.

IF and SWITCH

Definition: IF checks one condition. SWITCH checks many conditions in a clean way.

Simple Explanation: IF returns one result if true and another if false. SWITCH(TRUE(), ...) is a neat way to write many IFs.

Example: Label sales as High, Medium or Low.

```
Status =
SWITCH(TRUE(),
    [Total Sales] >= 100000, "High",
    [Total Sales] >= 50000,  "Medium",
    "Low")
```

Technique (Step-by-Step): 1. Write the condition first. 2. Use IF for two outcomes. 3. Use SWITCH(TRUE(), ...) for three or more. 4. Add a final default value at the end.

Interview Answer: “I use IF for simple yes or no checks. When there are many levels, like High, Medium and Low, I use SWITCH so the formula stays easy to read.”

Follow-up Qs:
- **Q:** Why SWITCH instead of nested IF? **A:** It is easier to read and fix.
- **Q:** What is the last value in SWITCH? **A:** The default, used when nothing else matches.

Common Mistake: Writing the conditions in the wrong order.

Related: Similar to Excel IF and SQL CASE WHEN.

DIVIDE

Definition: DIVIDE divides two numbers and handles divide-by-zero safely.

Simple Explanation: If the bottom number is zero, normal division gives an error. DIVIDE gives a blank, or a value you choose.

Example: Profit margin = Profit divided by Sales.

```
Profit Margin % =
DIVIDE([Total Profit], [Total Sales], 0)
```

Technique (Step-by-Step): 1. Write DIVIDE(then the top number, a comma, the bottom number. 2. Optionally add a third value to show when dividing by zero. 3. Format the measure as a Percentage.

Interview Answer: “I always use DIVIDE instead of the slash sign, because it handles zero safely and does not break my report.”

Follow-up Qs:
- **Q:** What does DIVIDE return for division by zero? **A:** Blank, or the alternate value you give.

- **Q:** Why not use the / sign? **A:** It can give errors.

Common Mistake: Using / and getting errors in the report.

Related: Used in growth and margin measures.

CALCULATE

Definition: CALCULATE changes the filters and then calculates a value. It is the most important DAX function.

Simple Explanation: You give it a measure and one or more filters. It applies those filters and gives the result.

Example: Sales for only the 'West' region, whatever the slicer says.

```
West Sales =
CALCULATE([Total Sales], Region[Name] = "West")

Big Orders =
CALCULATE([Total Sales], Sales[Amount] > 1000)
```

Technique (Step-by-Step): 1. Write CALCULATE(and then the measure you want. 2. Add a comma and a filter, like Region[Name] = "West". 3. Add more filters if needed. 4. Test it in a table next to the normal measure.

Interview Answer: “CALCULATE lets me change the filters for a measure. For example, I use it to show sales only for one region, or only for orders above a certain size.”

Follow-up Qs:
- **Q:** What does CALCULATE do in one line? **A:** It runs a calculation after changing the filters.
- **Q:** Can you use more than one filter? **A:** Yes. All filters are applied together.

Common Mistake: Forgetting that CALCULATE overrides existing filters on the same column.

Related: Leads to FILTER and ALL.

FILTER, ALL and ALLEXCEPT

Definition: FILTER returns the rows that meet a condition. ALL removes filters. ALLEXCEPT removes all filters except the ones you name.

Simple Explanation: ALL is used when you want to ignore slicers, like for a grand total. FILTER is used for conditions that need a full table check.

Example: Percentage of total: sales divided by sales for all products.

```
% of Total =
DIVIDE([Total Sales],
    CALCULATE([Total Sales], ALL(Product)))

Sales of Big Orders =
CALCULATE([Total Sales],
    FILTER(Sales, Sales[Amount] > 1000))
```

Technique (Step-by-Step): 1. Use ALL(table) inside CALCULATE to remove filters from that table. 2. Use ALLEXCEPT to keep only the filters you want. 3. Use FILTER when your condition uses a measure or many columns.

Interview Answer: “I use ALL when I need a total that ignores the slicers, for example to work out percentage of total. I use FILTER when the condition is more than a simple column check.”

Follow-up Qs:
- **Q:** ALL vs ALLEXCEPT? **A:** ALL removes every filter on a table. ALLEXCEPT removes all except the columns you name.
- **Q:** When do you need FILTER? **A:** When the condition cannot be written as a simple column test.

Common Mistake: Using ALL on the wrong table, so the percentage is wrong.

Related: Used in % of total (Level 6).

VALUES and SELECTEDVALUE

Definition: VALUES returns the unique values left in a column. SELECTEDVALUE returns the value if only one is selected.

Simple Explanation: SELECTEDVALUE is handy for dynamic titles. If a user picks one region in a slicer, it shows that region name.

Example: Chart title that says 'Sales for West'.

```
Title =
"Sales for " & SELECTEDVALUE(Region[Name], "All Regions")
```

Technique (Step-by-Step): 1. Create a new measure with SELECTEDVALUE. 2. Give an alternate result for when many or no values are selected. 3. Use it in a card, or in a title via Conditional formatting (fx).

Interview Answer: “SELECTEDVALUE gives me the one value picked in a slicer. I use it to make titles that change, like 'Sales for West', or 'All Regions' when nothing is picked.”

Follow-up Qs:
- **Q:** What does SELECTEDVALUE return if two are selected? **A:** Blank, or the alternate value you provided.
- **Q:** What does VALUES return? **A:** A list of the unique values in the current filter.

Common Mistake: Forgetting the alternate value, so the title looks empty.

Related: Related to slicers.

Variables (VAR and RETURN)

Definition: A variable stores a value inside a measure so you can reuse it.

Simple Explanation: It makes formulas shorter, easier to read and sometimes faster. You write VAR to create it and RETURN to give the final result.

Example: Growth % using this year and last year values.

```
Growth % =
VAR Current = [Total Sales]
VAR Previous = [Sales LY]
RETURN
    DIVIDE(Current - Previous, Previous)
```

Technique (Step-by-Step): 1. Write VAR and give the variable a name and a value. 2. Add more VAR lines if needed. 3. Finish with RETURN and the final formula that uses the variables.

Interview Answer: “I use VAR and RETURN to break a formula into small steps with clear names. It makes it easier for me and others to read and fix.”

Follow-up Qs:
- **Q:** Why use variables? **A:** Readability, and the value is calculated once.
- **Q:** Can you use many variables? **A:** Yes.

Common Mistake: Writing one huge formula that no one can read.

Related: Good practice for all DAX.

RELATED and Iterators (SUMX)

Definition: RELATED brings a value from a related table into the current row. SUMX goes row by row and adds up the result.

Simple Explanation: Use RELATED to pull Category into the Sales table. Use SUMX when you need to multiply before adding, like Quantity times Price.

Example: Revenue = Quantity times Price for every row, then summed.

```
Category = RELATED(Product[Category])        -- calculated column

Revenue =
SUMX(Sales, Sales[Qty] * Sales[Price])
```

Technique (Step-by-Step): 1. Use RELATED in a calculated column on the many side to get a value from the one side. 2. Use SUMX(table, expression) when each row needs a calculation before summing. 3. Check against a sample row by hand.

Interview Answer: “RELATED pulls a value from another table, like a lookup. SUMX works row by row, for example Quantity times Price, and then adds the results.”

Follow-up Qs:

- **Q:** SUM vs SUMX? **A:** SUM adds one column. SUMX calculates for each row first, then adds.
- **Q:** Does RELATED need a relationship? **A:** Yes.

Common Mistake: Using SUM(Qty) * SUM(Price), which gives the wrong answer.

Related: Similar to Excel VLOOKUP and SUMPRODUCT.

Level 5 - Quick Revision

Function	Simple meaning
Measure	Calculates with filters; use it most
Calculated column	Row by row; stored in the model
CALCULATE	Change filters, then calculate
ALL	Remove filters
FILTER	Rows that meet a condition
DIVIDE	Safe division
SWITCH	Many IFs in a clean way
VAR / RETURN	Store values for reuse
RELATED	Lookup from another table
SUMX	Row by row, then add

Key rule: Use measures for numbers and columns only when you must.

Top Interview Questions: (1) Calculated column vs measure. (2) What is CALCULATE? (3) What is filter context? (4) SUM vs SUMX.

Practical Scenarios: (1) Total sales that react to slicers - measure. (2) Sales for one region only - CALCULATE. (3) Percentage of total - CALCULATE + ALL + DIVIDE.

LEVEL 6 - DAX TIME INTELLIGENCE AND RANKING

Year to Date (TOTALYTD)

Definition: TOTALYTD gives the total from the start of the year up to the current date.

Simple Explanation: It needs a proper date table. It is used in almost every sales and finance report.

Example: Sales from 1 January up to the month shown.

```
Sales YTD =
TOTALYTD([Total Sales], 'Date'[Date])
```

Technique (Step-by-Step): 1. Make sure you have a date table marked as Date table. 2. Create a measure with TOTALYTD. 3. Put Month from the Date table on the axis of a chart. 4. Check that the value grows month by month.

Interview Answer: “TOTALYTD gives me the running total from the start of the year, month by month. It needs a proper date table to work correctly.”

Follow-up Qs:

- **Q:** What if my fiscal year starts in April? **A:** Add a third value: TOTALYTD([Total Sales], 'Date'[Date], "3/31").
- **Q:** Why does it not work sometimes? **A:** Usually the date table is missing or not marked.

Common Mistake: Using the date column from the fact table instead of the Date table.

Related: Needs a date table (Level 4).

Same Period Last Year and YoY Growth

Definition: SAMEPERIODLASTYEAR gives the same dates one year ago. Use it to compare this year with last year.

Simple Explanation: First make a Last Year measure. Then calculate the difference or percentage growth.

Example: Sales in March this year compared with March last year.

```
Sales LY =
CALCULATE([Total Sales],
    SAMEPERIODLASTYEAR('Date'[Date]))

YoY % =
DIVIDE([Total Sales] - [Sales LY], [Sales LY])
```

Technique (Step-by-Step): 1. Create the Sales LY measure using CALCULATE and SAMEPERIODLASTYEAR. 2. Create YoY % using DIVIDE. 3. Format as Percentage. 4. Show both in a table by Month to check.

Interview Answer: “I make a Last Year measure with SAMEPERIODLASTYEAR, then compare it to this year using DIVIDE to get YoY growth. This is one of the most common business questions.”

Follow-up Qs:

- **Q:** What does YoY mean? **A:** Year over year: this year compared to the same period last year.
- **Q:** Why DIVIDE? **A:** If last year is zero, it avoids errors.

Common Mistake: Using it without a proper date table.

Related: Uses DIVIDE and CALCULATE.

Month over Month (DATEADD)

Definition: DATEADD moves dates back or forward by a number of days, months, quarters or years.

Simple Explanation: Use it to compare this month with the previous month.

Example: Sales in April compared with March.

```
Sales Prev Month =
CALCULATE([Total Sales],
    DATEADD('Date'[Date], -1, MONTH))

MoM % =
DIVIDE([Total Sales] - [Sales Prev Month],
       [Sales Prev Month])
```

Technique (Step-by-Step): 1. Create the Previous Month measure using DATEADD with -1 and MONTH. 2. Create MoM % using DIVIDE. 3. Show them in a table by Month to confirm.

Interview Answer: “DATEADD lets me move back one month, so I can compare this month to last month. I use negative numbers to go back in time.”

Follow-up Qs:
- **Q:** How to go back one quarter? **A:** DATEADD('Date'[Date], -1, QUARTER).
- **Q:** Does DATEADD need a date table? **A:** Yes.

Common Mistake: Using it on a date column that has gaps.

Related: Related to SAMEPERIODLASTYEAR.

Running Total

Definition: A running total adds up values one after another over time.

Simple Explanation: Each point on the chart shows everything added up so far.

Example: Cumulative sales through the year.

```
Running Total =
CALCULATE([Total Sales],
    FILTER(ALL('Date'),
        'Date'[Date] <= MAX('Date'[Date])))
```

Technique (Step-by-Step): 1. Write CALCULATE with your measure. 2. Use FILTER over ALL('Date') to keep all dates up to the current one. 3. Use MAX of the date to know the current point. 4. Show it in a line chart.

Interview Answer: “For a running total, I remove the date filter and then keep all dates up to the current date. That gives me cumulative sales so far.”

Follow-up Qs:
- **Q:** Why ALL on the Date table? **A:** To remove the current date filter and look at earlier dates too.
- **Q:** Other use? **A:** Cumulative customers or cumulative cost.

Common Mistake: Forgetting to use the Date table in the filter.

Related: Uses CALCULATE, FILTER and ALL.

Percentage of Total

Definition: It shows each item's share of the total.

Simple Explanation: Divide the item's sales by the total sales of everything, ignoring the item filter.

Example: Each product's share of overall sales.

```
% of Total =
DIVIDE([Total Sales],
    CALCULATE([Total Sales], ALL(Product)))
```

Technique (Step-by-Step): 1. Write DIVIDE. 2. On top, use the normal measure. 3. On the bottom, use CALCULATE with ALL on the table you are splitting by. 4. Format as Percentage.

Interview Answer: “I divide each item's sales by the total of all items. The total is calculated with ALL, so it ignores the product filter.”

Follow-up Qs:
- **Q:** ALL vs ALLSELECTED here? **A:** ALL ignores every filter on the table. ALLSELECTED keeps what the user selected in slicers.

- **Q:** Where is this used? **A:** Market share, category contribution and so on.

Common Mistake: Using ALL when the user expects slicers to still apply.

Related: Uses ALL (Level 5).

RANKX

Definition: RANKX gives a rank number, like 1, 2, 3, based on a value.

Simple Explanation: Use it to rank products, customers or stores by sales.

Example: Rank each product by total sales.

```
Product Rank =
RANKX(ALL(Product[ProductName]),
      [Total Sales], , DESC)
```

Technique (Step-by-Step): 1. Write RANKX and the list to rank (ALL of the product names). 2. Add the measure to rank by. 3. Add DESC for highest first. 4. Show it in a table next to the product and sales.

Interview Answer: “RANKX ranks items by a value. I use ALL on the product names so every product is ranked, and DESC so the best seller is number 1.”

Follow-up Qs:
- **Q:** What if two products tie? **A:** They get the same rank by default.
- **Q:** Why ALL inside RANKX? **A:** So that the rank compares against all products, not only the visible row.

Common Mistake: Leaving out ALL, so every row gets rank 1.

Related: Similar to SQL RANK.

Top N (TOPN and Top N filter)

Definition: Top N shows only the best few items, like the top 5 products.

Simple Explanation: The easy way is the Top N option in the Filters pane. The DAX way is the TOPN function.

Example: Show the top 5 customers by sales.

```
Top 5 Sales =
CALCULATE([Total Sales],
    TOPN(5, ALL(Customer[Name]), [Total Sales]))
```

Technique (Step-by-Step): 1. Easy way: select the visual, open Filters, choose the field, set Filter type to Top N, choose 5 and the measure to rank by, then Apply filter. 2. DAX way: use TOPN inside CALCULATE as shown.

Interview Answer: “For a quick top 5, I use the Top N filter in the Filters pane. When I need the total of the top 5 in a measure, I use TOPN in DAX.”

Follow-up Qs:
- **Q:** Top N filter vs TOPN? **A:** The filter is quick and visual-level. TOPN is inside a measure.
- **Q:** How to show bottom 5? **A:** Choose Bottom in the Top N filter.

Common Mistake: Choosing the wrong measure to rank by.

Related: Related to RANKX.

Level 6 - Quick Revision

Function	Simple meaning
TOTALYTD	Total from start of year to date
SAMEPERIODLASTYEAR	Same dates last year
DATEADD	Move dates back or forward
Running total	Cumulative value so far
% of total	Share of the whole (use ALL)
RANKX	Rank items
TOPN	Best N items

Key rule: Time functions need a proper, marked Date table.

Top Interview Questions: (1) How do you calculate YoY growth? (2) Why do you need a date table? (3) How do you make a running total? (4) Rank vs Top N.

Practical Scenarios: (1) CEO wants this year vs last year - SAMEPERIODLASTYEAR. (2) Show top 5 products - Top N filter. (3) Show each category's share of sales - % of total.

LEVEL 7 - VISUALS AND REPORT DESIGN

Choosing the Right Chart

Definition: The right chart depends on the question you want to answer.

Simple Explanation: Bar or column to compare items. Line to show change over time. Pie or donut for a few parts of a whole. Table for exact numbers. Card for one big number. Map for places. Scatter to see if two numbers are related.

Example: Sales by region: bar chart. Sales by month: line chart. Total sales today: card.

Technique (Step-by-Step): 1. Write down the question first. 2. Comparing items? Use a bar. 3. Trend over time? Use a line. 4. Share of total with few parts? Use a donut. 5. Need exact values? Use a table.

Interview Answer: "I first think about the question the user wants to answer, then pick the chart. Bars for comparing, lines for time, cards for key numbers. I avoid pie charts with many slices because they are hard to read."

Follow-up Qs:
- **Q:** When do you avoid a pie chart? **A:** When there are more than 4 or 5 slices.
- **Q:** Best chart for trend? **A:** A line chart.

Common Mistake: Using fancy charts that look nice but are hard to read.

Related: Leads to other visuals.

Table vs Matrix

Definition: A table shows rows and columns as they are. A matrix works like a pivot table, with fields on rows and columns and totals.

Simple Explanation: Use a matrix when you want to compare across two things, like Product by Month. Use a table for a simple list.

Example: A matrix with Region on rows, Month on columns and Sales as values.

Technique (Step-by-Step): 1. Click the Matrix visual. 2. Drag Region to Rows. 3. Drag Month to Columns. 4. Drag Sales to Values. 5. Turn on Expand to go down hierarchy levels.

Interview Answer: "I use a table for a plain list and a matrix when I want a pivot-style view, like sales by region and by month."

Follow-up Qs:
- **Q:** Which one can expand and collapse? **A:** The matrix.
- **Q:** Is matrix like an Excel pivot table? **A:** Yes, it is very similar.

Common Mistake: Using a table when a matrix would show the pattern much better.

Related: Similar to Excel PivotTable.

Card and KPI Visuals

Definition: A card shows one number. A KPI visual shows a value against a target, with a trend.

Simple Explanation: These go at the top of a dashboard to show the main numbers at a glance.

Example: Total Sales, Total Profit and Profit Margin as cards at the top.

Technique (Step-by-Step): 1. Click the Card visual and drag a measure into Fields. 2. For a KPI: pick the KPI visual, add the value, a trend axis (like Month) and a target. 3. Format the colors for good and bad.

Interview Answer: "I put the main numbers in cards at the top of my report, so the manager sees the key figures in 5 seconds. For targets I use a KPI visual."

Follow-up Qs:
- **Q:** What do you need for a KPI visual? **A:** A value, a target and a trend axis.
- **Q:** How many cards are good at the top? **A:** About 3 to 5.

Common Mistake: Putting too many cards on a page.

Related: Related to KPIs (Level 10).

Slicers

Definition: A slicer is a filter that sits on the report page and lets users click to pick values.

Simple Explanation: Slicers can be a list, a dropdown, buttons, or a date range slider. You can sync a slicer across many pages.

Example: Region and Year slicers on a sales page.

Technique (Step-by-Step): 1. Click the Slicer visual. 2. Drag a field (Region) into it. 3. Format as List, Dropdown or Between. 4. To use on many pages: View tab, Sync slicers, tick the pages. 5. Use Edit interactions to stop a slicer from affecting some visuals.

Interview Answer: “Slicers let users filter the page themselves. I keep them simple and use sync slicers so the choice follows the user to other pages.”

Follow-up Qs:
- **Q:** What is Sync slicers? **A:** It makes the same slicer work on many pages.
- **Q:** How to stop a slicer from changing a visual? **A:** Use Edit interactions.

Common Mistake: Adding too many slicers, which makes the report confusing.

Related: Related to filters.

Filters: Visual, Page and Report Level

Definition: Filters in the Filters pane limit data at three levels: one visual, one page, or the whole report.

Simple Explanation: Visual filter affects one chart. Page filter affects all charts on the page. Report filter affects every page. You can also set filter types: Basic, Advanced, and Top N.

Example: A report filter to show only the current year, a visual filter for the top 5 products.

Technique (Step-by-Step): 1. Select a visual. 2. Open the Filters pane on the right. 3. Drag a field to Filters on this visual, Filters on this page or Filters on all pages. 4. Choose values or a condition.

Interview Answer: “I use report-level filters for rules that apply everywhere, like 'only the current year', and visual-level filters for one chart, like the top 5.”

Follow-up Qs:
- **Q:** Slicer vs filter? **A:** A slicer is visible and used by the viewer. A filter in the pane is set by the report builder.
- **Q:** Can you lock a filter? **A:** Yes. You can lock or hide filters.

Common Mistake: Setting a visual filter and forgetting it is there.

Related: Related to slicers.

Conditional Formatting

Definition: Conditional formatting changes colors or adds icons and data bars based on values.

Simple Explanation: It helps people see good and bad numbers quickly, like red for below target and green for above.

Example: Profit Margin below 10 percent in red.

Technique (Step-by-Step): 1. Select the table or matrix. 2. In the Visualizations pane, open the field drop-down. 3. Choose Conditional formatting, then Background color, Font color, Data bars or Icons. 4. Set the rules, for example Rules based on a measure. 5. Click OK.

Interview Answer: “I use conditional formatting to highlight problems, like red for sales below target. People notice it faster than reading numbers.”

Follow-up Qs:
- **Q:** What types are there? **A:** Background color, font color, data bars and icons.
- **Q:** Can it use a measure? **A:** Yes, with 'Rules' or 'Field value' formatting.

Common Mistake: Using too many colors, so nothing stands out.

Related: Related to design tips.

Tooltips

Definition: A tooltip is a small box that appears when you hover over a visual.

Simple Explanation: You can add extra fields to the default tooltip. You can also build a report page tooltip that shows a small chart.

Example: Hover over a bar and see profit and order count.

Technique (Step-by-Step): 1. Select the visual. 2. In the Visualizations pane, open the Format section and look for Tooltips. 3. Drag extra fields into Tooltips. 4. For a page tooltip: make a new page, in Page information turn on Allow use as tooltip, and set the tooltip in the main visual.

Interview Answer: “Tooltips let me show extra details on hover, so the main chart stays clean. For some reports I build a small chart as a tooltip page.”

Follow-up Qs:
- **Q:** Why use tooltips? **A:** To add detail without making the page crowded.
- **Q:** Can a tooltip be a chart? **A:** Yes, using a report page tooltip.

Common Mistake: Putting key information only in tooltips, where people might never look.

Related: Related to design tips.

Drill Down vs Drill Through

Definition: Drill down goes deeper inside the same visual, like Year to Month. Drill through jumps to another page with details about the item you picked.

Simple Explanation: Drill down needs a hierarchy. Drill through needs a detail page with the field in 'Drill through' filters.

Example: Right-click a product in the summary page and go to the Product Details page.

Technique (Step-by-Step): 1. For drill down: add a hierarchy to the visual and turn on the drill mode arrow. 2. For drill through: create a new detail page. 3. Drag the field (Product) into Drill through on that page. 4. On the summary page, right-click an item, then Drill through, then the page name.

Interview Answer: “Drill down goes deeper in the same chart, like Year to Quarter to Month. Drill through takes the user to a separate detail page already filtered for the item they picked.”

Follow-up Qs:
- **Q:** Drill down vs drill through? **A:** Down: same visual. Through: another page.
- **Q:** What do you need for drill through? **A:** A page with the drill through field set.

Common Mistake: Making a drill through page but forgetting a Back button.

Related: Related to hierarchies (Level 4).

Bookmarks and Buttons

Definition: A bookmark saves the current state of a page (filters, visible visuals). Buttons can use bookmarks to switch views.

Simple Explanation: A common use: a toggle button that switches between a chart and a table. Another use: a reset button to clear all filters.

Example: Button 'Chart view' and 'Table view' on one page.

Technique (Step-by-Step): 1. Set the page the way you want it. 2. View tab, Bookmarks, Add. 3. Insert a Button from the Insert tab. 4. In Format, Action, choose Bookmark and the saved one. 5. Test it in Reading view or with Ctrl+click.

Interview Answer: “I use bookmarks and buttons to make reports more interactive, like switching between a chart view and a table view, or a reset-filters button.”

Follow-up Qs:
- **Q:** What does a bookmark save? **A:** Filters, slicer values and which visuals are visible.
- **Q:** What do you use the Selection pane for? **A:** To show or hide visuals for each bookmark.

Common Mistake: Creating bookmarks before the page is final, then changing the layout.

Related: Related to navigation.

Themes and Good Design Tips

Definition: A theme sets the colors and fonts for the whole report. Good design makes reports easy to read.

Simple Explanation: Use few colors, keep spacing even, put the most important numbers at the top left, add clear titles and avoid clutter.

Example: A sales report with 3 cards on top, one trend chart, one bar chart and two slicers.

Technique (Step-by-Step): 1. View tab, Themes, pick one or import a JSON theme from your company. 2. Put key numbers on top. 3. Align visuals using Format, Align. 4. Use the same colors for the same meaning. 5. Remove anything that is not needed.

Interview Answer: “I keep reports clean: key numbers on top, one or two charts, few colors and clear titles. If a visual does not help answer the question, I remove it.”

Follow-up Qs:
- **Q:** What is the best layout? **A:** Key numbers at the top, trends in the middle, details at the bottom.
- **Q:** How many visuals per page? **A:** Usually about 5 to 8.

Common Mistake: Filling the page with too many charts and bright colors.

Related: Summary of this level.

Level 7 - Quick Revision

Feature	Simple meaning
Bar / Line / Card	Compare / trend / one number
Matrix	Pivot-style view
Slicer	Click filter on the page
Visual / Page / Report filter	Three filter levels
Conditional formatting	Color by value
Drill down	Deeper in same visual
Drill through	Jump to a detail page
Bookmark	Saves a page state

Key rule: Pick the chart for the question, and keep the page clean.

Top Interview Questions: (1) Drill down vs drill through. (2) Slicer vs filter. (3) Table vs matrix. (4) How do you choose a chart?

Practical Scenarios: (1) Manager wants to compare regions - bar chart. (2) User wants details of one product - drill through page. (3) Highlight sales below target - conditional formatting.

LEVEL 8 - POWER BI SERVICE AND SHARING

Workspaces and Roles

Definition: A workspace is a shared folder in the Power BI Service where reports, dashboards and models live.

Simple Explanation: People in a workspace get a role: Admin, Member, Contributor or Viewer. The role decides what they can do.

Example: A 'Sales Team' workspace where analysts publish and managers view.

Technique (Step-by-Step): 1. In the Service, click Workspaces, then Create a workspace. 2. Name it and save. 3. Click Access to add people. 4. Give each person a role (Viewer for readers, Contributor or Member for builders).

Interview Answer: “Workspaces are shared spaces for a team. I give viewers the Viewer role, and builders get Member or Contributor, so each person can only do what they need.”

Follow-up Qs:
- **Q:** Which role can only view? **A:** Viewer.
- **Q:** What is My workspace? **A:** A private space for your own work.

Common Mistake: Giving everyone Admin access.

Related: Related to licenses (Level 1).

Publish a Report

Definition: Publishing sends your report from Desktop to the Power BI Service.

Simple Explanation: After publishing, the report and its semantic model appear in the workspace you chose.

Example: Publish the monthly sales report to the Sales workspace.

Technique (Step-by-Step): 1. In Desktop, Home tab, Publish. 2. Sign in if asked. 3. Select the workspace. 4. Click Select. 5. Open the link to see the report in the Service.

Interview Answer: “I click Publish in Desktop, choose the right workspace, and then check the report in the Service to be sure everything looks right.”

Follow-up Qs:
- **Q:** What if you publish again? **A:** You can replace the old report with the new one.
- **Q:** Where does the model go? **A:** It is published with the report as a semantic model.

Common Mistake: Publishing to the wrong workspace.

Related: Leads to refresh.

Report vs Dashboard

Definition: A report is one or more pages built from one model. A dashboard is a single page in the Service made of tiles pinned from different reports.

Simple Explanation: Reports are made in Desktop or the Service. Dashboards are made only in the Service. A dashboard gives one view of the key numbers from many reports.

Example: A CEO dashboard with tiles from the sales, finance and HR reports.

Technique (Step-by-Step): 1. Open a report in the Service. 2. Hover over a visual and click the pin icon. 3. Choose a new or existing dashboard. 4. Repeat for visuals from other reports. 5. Open the dashboard from the workspace.

Interview Answer: “A report has many pages and comes from one model. A dashboard is one page of pinned tiles that can come from many reports. Managers like dashboards for quick checks.”

Follow-up Qs:
- **Q:** Where do you build a dashboard? **A:** Only in the Power BI Service.
- **Q:** Can a dashboard use many reports? **A:** Yes.

Common Mistake: Calling every report a dashboard.

Related: Common interview question.

Scheduled Refresh and Gateway

Definition: Scheduled refresh updates the data in the Service automatically. A gateway is needed if the data is inside your company network.

Simple Explanation: Cloud sources like SharePoint Online often do not need a gateway. Files on a local drive or on-premises SQL Server do need one. A Pro license allows up to 8 scheduled refreshes a day.

Example: Sales data refreshes every morning at 7 AM.

Technique (Step-by-Step): 1. In the Service, find the semantic model, click the three dots, then Settings. 2. Open Data source credentials and sign in. 3. If needed, connect the gateway. 4. Open Refresh, turn on the schedule and add the times. 5. Click Apply.

Interview Answer: “I set up a scheduled refresh so the report updates automatically. If the data is on our local network, I use a gateway so the Service can reach it.”

Follow-up Qs:
- **Q:** When do you need a gateway? **A:** When data is on-premises or on a local machine.
- **Q:** What if refresh fails? **A:** Check credentials, gateway status, and whether the source changed.

Common Mistake: Not setting credentials, so the refresh fails every night.

Related: Related to Import mode (Level 2).

Sharing and Apps

Definition: You can share one report with a link, or package many reports into an app for a large group.

Simple Explanation: Sharing is fine for a few people. An app is better for many users, because it is neat, read-only and can have its own navigation.

Example: Send the sales report to a manager, and publish an app for 100 sales staff.

Technique (Step-by-Step): 1. To share: open the report, click Share, and enter names or emails. 2. To make an app: in the workspace, click Create app. 3. Add a name, content and audience. 4. Publish the app and share the link.

Interview Answer: “For a few people I use Share. For a bigger audience I publish an app, so users get a clean, read-only version.”

Follow-up Qs:
- **Q:** Share vs app? **A:** Share is for a few people. An app is for a large audience.
- **Q:** Do viewers need a license? **A:** Usually Pro, unless the content is in Premium or Fabric capacity.

Common Mistake: Sharing the report with edit access instead of view access.

Related: Related to workspaces and licenses.

Row-Level Security (RLS)

Definition: RLS makes sure each person sees only the data they are allowed to see.

Simple Explanation: You create roles with a DAX filter in Desktop, publish, and then add users to the role in the Service.

Example: A regional manager sees only their own region's sales.

```
[Region] = "West"
```

Technique (Step-by-Step): 1. Desktop: Modeling tab, Manage roles. 2. Create a role and pick the table. 3. Add a filter like [Region] = "West". 4. Test with View as. 5. Publish. 6. In the Service, open the model's Security and add people to the role.

Interview Answer: “With RLS, each user sees only their own data. I create roles with a simple filter, test them using View as, and then add the right people in the Service.”

Follow-up Qs:
- **Q:** Static vs dynamic RLS? **A:** Static uses a fixed value per role. Dynamic uses the user's login (USERPRINCIPALNAME) to filter.

- **Q:** Do workspace Admins and Members bypass RLS? **A:** Yes. RLS applies to Viewers.

Common Mistake: Not testing the role before publishing.

Related: Related to workspaces.

Subscriptions and Alerts

Definition: A subscription emails a snapshot of a report on a schedule. An alert notifies you when a number crosses a limit.

Simple Explanation: Subscriptions are good for people who never open Power BI. Alerts work on dashboard cards, gauges and KPI tiles.

Example: A daily 8 AM sales email, and an alert if sales fall below 1 lakh.

Technique (Step-by-Step): 1. For a subscription: open the report, click Subscribe, set the time and people. 2. For an alert: on a dashboard tile, click the three dots, then Manage alerts. 3. Set the rule and click Save.

Interview Answer: "I set up subscriptions so managers get the report by email on schedule, and alerts so I know right away when a key number goes beyond a limit."

Follow-up Qs:
- **Q:** Where do alerts work? **A:** On dashboard tiles such as cards, KPIs and gauges.
- **Q:** Who is a subscription useful for? **A:** People who do not log into Power BI.

Common Mistake: Forgetting that the subscription needs the right license and access.

Related: Related to MIS reporting (Level 10).

Level 8 - Quick Revision

Item	Simple meaning
Workspace	Shared space for a team
Publish	Send report from Desktop to Service
Dashboard	One page of pinned tiles
Gateway	Bridge to on-premises data
Scheduled refresh	Auto-update data
App	Packaged reports for many users
RLS	Each person sees their own data
Subscription	Scheduled email of a report

Key rule: Set the refresh, set the credentials and test RLS before sharing.

Top Interview Questions: (1) Report vs dashboard. (2) What is a gateway? (3) What is RLS and how do you set it up? (4) How do you share reports with many people?

Practical Scenarios: (1) Data stops updating - check refresh, credentials and gateway. (2) Regional managers must see only their region - RLS. (3) 200 people need the report - publish an app.

LEVEL 9 - PERFORMANCE AND GOOD PRACTICE

Why Reports Become Slow

Definition: A report is slow when the model is too big or too messy, or the formulas are heavy.

Simple Explanation: Common causes: too many columns, too many rows, many visuals on one page, bad relationships and heavy DAX.

Example: A report with 50 visuals on one page that takes 30 seconds to open.

Technique (Step-by-Step): 1. Open the report and see which page is slow. 2. Count visuals on that page. 3. Check the model size and number of columns. 4. Use Performance Analyzer to find the slowest visual.

Interview Answer: “When a report is slow, I first check the model size, the number of visuals on the page, and heavy formulas. Then I fix the biggest problem first.”

Follow-up Qs:

- **Q:** First thing to check? **A:** Use Performance Analyzer to see which visual is slow.
- **Q:** Does more visuals mean slower? **A:** Yes, each visual runs its own query.

Common Mistake: Guessing the problem without measuring.

Related: Leads to Performance Analyzer.

Performance Analyzer

Definition: Performance Analyzer is a Power BI Desktop tool that shows how long each visual takes to load.

Simple Explanation: It splits the time into DAX query, visual display and other. You can copy the query and test it in other tools.

Example: Find that one matrix takes 8 seconds while the others take 1 second.

Technique (Step-by-Step): 1. In Desktop, go to the View tab. 2. Click Performance analyzer. 3. Click Start recording. 4. Click Refresh visuals. 5. Read the time for each visual and fix the slowest one.

Interview Answer: “I use Performance Analyzer to see which visual is slow and whether the time is spent in DAX or in drawing the visual. Then I fix that one first.”

Follow-up Qs:

- **Q:** Where is it? **A:** View tab, Performance analyzer.
- **Q:** What does it show? **A:** Time taken by each visual, split into DAX query and display.

Common Mistake: Not measuring before optimizing.

Related: Related to reducing model size.

Reduce Data Size

Definition: Smaller models are faster. Remove what you do not need.

Simple Explanation: Delete unused columns and rows, keep only the history you need, avoid text columns with many unique values, and turn off auto date/time for each date column.

Example: Remove 20 unused columns and keep only the last 3 years of data.

Technique (Step-by-Step): 1. In Power Query, remove columns you do not need. 2. Filter rows to the period required. 3. Change types to the smallest suitable one. 4. File, Options, Data Load, turn off Auto date/time. 5. Use your own Date table.

Interview Answer: “I keep only the columns and rows I need. Fewer columns and rows make the model smaller, so the report loads and refreshes faster.”

Follow-up Qs:

- **Q:** Which columns hurt size the most? **A:** Long text and ID columns with many unique values.
- **Q:** What about Auto date/time? **A:** It creates hidden date tables for every date column, so a proper Date table is better.

Common Mistake: Loading the whole database 'just in case'.

Related: Related to Power Query (Level 3).

Measures over Columns, and Careful Relationships

Definition: Prefer measures to calculated columns, and keep relationships simple.

Simple Explanation: Calculated columns take memory. Both-direction filters and many-to-many relationships can slow things down.

Example: Replace a calculated column with a measure, and change a Both filter to Single.

Technique (Step-by-Step): 1. List your calculated columns. 2. Ask whether each can be a measure. 3. Check relationships in Model view. 4. Change Both to Single when possible. 5. Re-test using Performance Analyzer.

Interview Answer: “I use measures instead of calculated columns when I can, and I keep relationships on single direction. This keeps the model light and quick.”

Follow-up Qs:
- **Q:** Why are measures lighter? **A:** They are not stored. They are calculated only when needed.
- **Q:** Why avoid many-to-many? **A:** It is slower and can give confusing results.

Common Mistake: Creating a calculated column for every small calculation.

Related: Related to DAX (Level 5).

Incremental Refresh

Definition: Incremental refresh refreshes only the new or changed data, not the whole table.

Simple Explanation: It is useful for large tables. Old data stays as it is, and only recent data refreshes. It needs Pro or higher and two date/time parameters named RangeStart and RangeEnd.

Example: A table with 5 years of sales refreshes only the last 7 days.

Technique (Step-by-Step): 1. In Power Query, create RangeStart and RangeEnd parameters. 2. Filter your date column using them. 3. Right-click the table in Desktop, choose Incremental refresh. 4. Set how much history to store and how much to refresh. 5. Publish.

Interview Answer: “For big tables, I use incremental refresh so only the recent data is refreshed. It saves time and resources.”

Follow-up Qs:
- **Q:** What parameters are needed? **A:** RangeStart and RangeEnd.
- **Q:** Why use it? **A:** Faster refresh for large data.

Common Mistake: Using it on small tables where a normal refresh is fine.

Related: Related to refresh (Level 8).

Naming, Documentation and Testing

Definition: Good habits that make your work easy to understand and trust.

Simple Explanation: Use clear names for tables, columns and measures. Group measures in a folder. Add descriptions. Check your numbers against the source before sharing.

Example: A measure named 'Total Sales' in a folder 'Sales Measures' with a short description.

Technique (Step-by-Step): 1. Rename columns and tables clearly. 2. Create a table to hold measures, or use display folders. 3. Add a description in the Properties pane. 4. Compare totals with Excel or SQL. 5. Ask a colleague to review.

Interview Answer: “I give things clear names and check my totals against the source data before I share. That way people can trust the report.”

Follow-up Qs:
- **Q:** How do you check your report is correct? **A:** Compare totals with the source, and test with a few rows by hand.
- **Q:** Why add descriptions? **A:** So others know what a measure means.

Common Mistake: Using names like Measure 1 and Table 2.

Related: Good habit for all levels.

Level 9 - Quick Revision

Tip	Simple meaning
Performance Analyzer	Find the slowest visual
Remove unused columns/rows	Smaller model, faster report
Use measures	Lighter than columns
Keep Single direction	Avoid Both unless needed
Incremental refresh	Refresh only new data
Clear names and testing	Trust and easy to maintain

Key rule: Measure first, then fix the biggest problem.

Top Interview Questions: (1) How do you make a slow report faster? (2) What is Performance Analyzer? (3) What is incremental refresh?

Practical Scenarios: (1) Report takes 40 seconds to open - Performance Analyzer, reduce visuals and data. (2) Refresh takes 2 hours - incremental refresh. (3) Colleague cannot understand your measures - naming and descriptions.

LEVEL 10 - ANALYST AND BUSINESS SKILLS

KPIs and Metrics

Definition: A KPI (Key Performance Indicator) is a number that shows how well something is going against a goal.

Simple Explanation: A metric is any number you measure. A KPI is a metric that matters most to the business. Good KPIs are clear, measurable and have a target.

Example: Monthly sales vs target, customer churn rate, average delivery time.

```
Target Achieved % =
DIVIDE([Total Sales], [Sales Target])
```

Technique (Step-by-Step): 1. Ask the business: what does success look like? 2. Pick 3 to 5 main KPIs. 3. Agree on the exact formula for each. 4. Add a target. 5. Show them at the top of the report.

Interview Answer: “A KPI is a key number tied to a goal, like monthly sales against target. I always agree on the exact formula with the business, so everyone reads the number the same way.”

Follow-up Qs:
- **Q:** KPI vs metric? **A:** A KPI is a metric tied to an important goal.
- **Q:** How many KPIs on a page? **A:** Usually 3 to 5.

Common Mistake: Showing many numbers without saying which ones matter.

Related: Related to cards and KPI visuals (Level 7).

Requirement Gathering

Definition: Requirement gathering means asking the user what they really need before building anything.

Simple Explanation: Ask: who will use it, what decisions will they take, what numbers do they need, how often, and where is the data. This avoids rework.

Example: A sales head says 'I need a dashboard'. You ask which questions it must answer.

Technique (Step-by-Step): 1. Meet the user. 2. Ask: what decision will this help you make? 3. List the KPIs and filters needed. 4. Ask where the data comes from and how often it changes. 5. Write it down and confirm with the user.

Interview Answer: “Before I open Power BI, I talk to the user and ask what questions the report must answer and what decisions they will take. I write it down and confirm, so I do not build the wrong thing.”

Follow-up Qs:
- **Q:** What if the user is not sure? **A:** Show a quick sketch or sample and ask for feedback.
- **Q:** Why write it down? **A:** To avoid misunderstandings later.

Common Mistake: Building the report first and asking questions later.

Related: Leads to wireframing.

Wireframe and Dashboard Planning

Definition: A wireframe is a rough sketch of the report before you build it.

Simple Explanation: It can be on paper or in PowerPoint. You decide where the cards, charts and slicers go. It saves time and shows the user the plan early.

Example: A sketch with 4 cards on top, a trend chart, a bar chart and two slicers on the left.

Technique (Step-by-Step): 1. Draw boxes on paper or a slide. 2. Put the most important numbers on top. 3. Show it to the user. 4. Change it based on feedback. 5. Then build in Power BI.

Interview Answer: “I make a quick sketch first and show it to the user. It is much easier to change a sketch than a finished report.”

Follow-up Qs:
- **Q:** Why wireframe? **A:** To agree on the layout early and avoid rework.

- **Q:** What tool do you use? **A:** Paper or PowerPoint is enough.

Common Mistake: Skipping the sketch and redoing the report three times.

Related: Related to design tips (Level 7).

Data Storytelling

Definition: Data storytelling means presenting data so people understand what it means and what to do.

Simple Explanation: Show the main point first, use clear titles, highlight what changed and why, and suggest an action.

Example: Title: 'Sales fell 12 percent in March because of the North region'.

Technique (Step-by-Step): 1. Find the main message in your data. 2. Put it in the title, not just 'Sales by Month'. 3. Highlight the key point with color. 4. Explain the reason. 5. Suggest the next step.

Interview Answer: “I try to tell a short story: what happened, why it happened, and what we should do. I put the main point in the title, so people understand right away.”

Follow-up Qs:
- **Q:** Why use insight titles? **A:** They tell the reader what to notice.
- **Q:** What makes a dashboard useful? **A:** It helps someone take a decision.

Common Mistake: Showing charts without explaining what matters.

Related: Related to design.

Common Analysis Types (Variance, Pareto, Funnel, Cohort)

Definition: These are common business analysis patterns you may be asked about.

Simple Explanation: Variance: actual vs target. Pareto (80/20): a few items give most of the result. Funnel: how many people move through each step. Cohort: how groups of customers behave over time.

Example: Actual vs budget variance; top 20 percent of products giving 80 percent of sales.

```
Variance = [Actual] - [Budget]
Variance % = DIVIDE([Variance], [Budget])
```

Technique (Step-by-Step): 1. Variance: make Actual and Budget measures, then subtract. 2. Pareto: rank items, add a cumulative percent, and show a bar plus a line. 3. Funnel: use the Funnel visual with the steps. 4. Cohort: group customers by first purchase month.

Interview Answer: “I use variance to compare actual with target. I use a Pareto view to find the few products that bring most of the sales. For steps in a process, I use a funnel.”

Follow-up Qs:
- **Q:** What is the 80/20 rule? **A:** About 80 percent of results come from 20 percent of causes.
- **Q:** What does a funnel show? **A:** How many users drop off at each step.

Common Mistake: Using fancy analysis when a simple comparison would do.

Related: Related to KPIs.

Product Analytics Metrics

Definition: Product analysts track how users use a product: active users, retention, churn and conversion.

Simple Explanation: DAU is daily active users. MAU is monthly active users. Churn is people who left. Conversion is people who took the step you want.

Example: DAU, MAU, retention and sign-up to purchase conversion.

```
Conversion % =
DIVIDE([Purchasers], [Visitors])

Churn % =
DIVIDE([Lost Customers], [Customers at Start])
```

Technique (Step-by-Step): 1. Define each metric clearly with the product team. 2. Build measures with DISTINCTCOUNT of user IDs. 3. Use a date table for daily and monthly views. 4. Show trends in line charts and steps in a funnel.

Interview Answer: “For product analysis I track users with measures like DISTINCTCOUNT of user ID for active users, then conversion and churn. I define each metric clearly with the product team first.”

Follow-up Qs:
- **Q:** DAU vs MAU? **A:** Daily active users and monthly active users.
- **Q:** What is retention? **A:** The share of users who come back after some time.

Common Mistake: Not agreeing on what counts as an 'active user'.

Related: Related to DISTINCTCOUNT (Level 5).

MIS Reporting

Definition: MIS (Management Information System) reports give managers regular numbers, daily, weekly or monthly.

Simple Explanation: They need to be accurate, on time and the same format each time. Power BI makes them automatic with scheduled refresh and subscriptions.

Example: A daily sales and attendance report sent every morning.

Technique (Step-by-Step): 1. Collect the same data sources each time. 2. Build the report once. 3. Set scheduled refresh. 4. Set a subscription so it goes by email. 5. Check totals against the source each day.

Interview Answer: “In MIS reporting, accuracy and timing matter most. I build the report once in Power BI, set it to refresh and email on a schedule, and always check the totals against the source.”

Follow-up Qs:
- **Q:** Biggest risk in MIS? **A:** Wrong or late numbers.
- **Q:** How do you reduce manual work? **A:** Power Query for cleaning, scheduled refresh and subscriptions.

Common Mistake: Copying and pasting data by hand every day.

Related: Related to subscriptions (Level 8).

Level 10 - Quick Revision

Skill	Simple meaning
KPI	Key number tied to a goal
Requirement gathering	Ask before you build
Wireframe	Sketch before building
Data storytelling	What happened, why, what next
Variance	Actual vs target
Pareto	80/20 rule
Funnel	Drop-off at each step
MIS report	Regular, accurate management report

Key rule: Understand the business question before you build.

Top Interview Questions: (1) How do you gather requirements? (2) What is a KPI? (3) What makes a good dashboard? (4) What is MIS reporting?

Practical Scenarios: (1) User asks for a dashboard with no details - ask questions and sketch a wireframe. (2) Team sends daily Excel reports by hand - automate with Power BI. (3) Product team wants conversion - funnel and conversion measure.

LEVEL 11 - PROJECT SCENARIOS AND INTERVIEW TALK

How to Explain Your Power BI Project

Definition: Interviewers almost always ask 'tell me about your project'. This is a simple way to answer.

Simple Explanation: Use this order: the business problem, the data, what you did, the result. Keep it short and use real numbers if you can.

Example: Sales dashboard: problem, Excel and SQL data, cleaning and model, result: report time cut from 2 days to 10 minutes.

Technique (Step-by-Step): 1. Say the problem: who needed what. 2. Say the data sources. 3. Say what you did in Power Query, modeling and DAX. 4. Say what visuals you made. 5. Say the result and who used it.

Interview Answer: “I built a sales dashboard for the sales team. The data came from SQL and Excel. I cleaned it in Power Query, built a star schema, wrote measures for sales, profit and growth, and published it. Now the team gets the report in minutes instead of waiting two days.”

Follow-up Qs:
- **Q:** What was the hardest part? **A:** Be honest. For example: messy data, or getting the date logic right.
- **Q:** What would you improve? **A:** Mention one simple thing, like adding more filters or faster refresh.

Common Mistake: Giving a long answer with no result at the end.

Related: Use your own real project details.

Sales Dashboard

Definition: A sales dashboard shows how much was sold, where, and how it changes over time.

Simple Explanation: Typical content: total sales, profit, orders, growth vs last year, sales by region and product, top customers.

Example: Cards for sales and profit, a line chart by month, a bar chart by region, a top 10 table.

```
Total Sales = SUM(Sales[Amount])
Sales LY = CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Date'[Date]))
YoY % = DIVIDE([Total Sales] - [Sales LY], [Sales LY])
```

Technique (Step-by-Step): 1. Get sales, product, customer and date data. 2. Clean in Power Query. 3. Build a star schema. 4. Write Total Sales, Profit, YoY %. 5. Build cards, trend, bar chart, top N. 6. Add slicers for Year and Region. 7. Publish and set refresh.

Interview Answer: “In my sales dashboard I showed total sales, profit and growth compared with last year at the top. Below that I had a monthly trend, sales by region, and the top products. Users could filter by year and region.”

Follow-up Qs:
- **Q:** Which measures did you write? **A:** Total Sales, Profit, Profit Margin, Sales LY and YoY percent.
- **Q:** What did managers use it for? **A:** To find weak regions and best products.

Common Mistake: Showing too many charts without a clear message.

Related: Uses Levels 3 to 7.

HR Dashboard

Definition: An HR dashboard shows employee numbers, hiring, attrition and attendance.

Simple Explanation: Typical content: headcount, new joiners, leavers, attrition rate, average tenure, department split.

Example: Cards for headcount and attrition, a trend of joiners and leavers, a bar by department.

```
Attrition % =
DIVIDE([Employees Left], [Average Headcount])
```

Technique (Step-by-Step): 1. Get employee master and attendance data. 2. Clean names, dates and departments. 3. Build the model with a Date table. 4. Write headcount, joiners, leavers and attrition measures. 5. Build visuals and add department and location slicers. 6. Use RLS so each manager sees only their team.

Interview Answer: “In the HR dashboard I tracked headcount, joiners, leavers and attrition rate by department. I used row-level security so managers only saw their own team.”

Follow-up Qs:
- **Q:** How did you calculate attrition? **A:** Employees who left divided by average headcount.
- **Q:** Why RLS here? **A:** HR data is private.

Common Mistake: Forgetting that HR data is sensitive and needs security.

Related: Uses RLS (Level 8).

MIS Daily Report Automation

Definition: Replacing a manual Excel report with an automatic Power BI report.

Simple Explanation: Before: someone copies data and builds charts by hand every day. After: Power Query cleans the data, refresh runs on schedule and a subscription emails the report.

Example: A daily operations report that used to take 2 hours.

Technique (Step-by-Step): 1. List what the manual report contains. 2. Connect to the same sources (folder or database). 3. Rebuild the cleaning in Power Query. 4. Make measures and visuals. 5. Set daily refresh and a subscription. 6. Compare with the old report for a week.

Interview Answer: “We had an Excel report that took two hours every day. I rebuilt it in Power BI with Power Query for cleaning and a scheduled refresh. Now the report arrives by email every morning and nobody touches it.”

Follow-up Qs:
- **Q:** How did you check it was correct? **A:** I compared it with the old Excel report for a week.
- **Q:** What was the benefit? **A:** Saved time and fewer mistakes.

Common Mistake: Switching over without comparing with the old report.

Related: Uses Levels 3 and 8.

Budget vs Actual (Finance)

Definition: A finance report compares what was planned with what really happened.

Simple Explanation: Typical content: actual vs budget, variance and variance percent by department or month, and colors for over or under budget.

Example: A matrix by department with Actual, Budget and Variance, with red for over budget.

```
Variance = [Actual] - [Budget]
Variance % = DIVIDE([Variance], [Budget])
```

Technique (Step-by-Step): 1. Get actual and budget tables. 2. Link both to common Date and Department tables. 3. Write Actual, Budget and Variance measures. 4. Show in a matrix. 5. Use conditional formatting for over and under budget.

Interview Answer: “In the finance report I compared actual with budget for each department. I used variance and variance percent, with red and green colors to show who is over or under budget.”

Follow-up Qs:
- **Q:** Why shared dimensions? **A:** So both Actual and Budget can be filtered by the same Date and Department.
- **Q:** What visual did you use? **A:** A matrix with conditional formatting.

Common Mistake: Linking Actual and Budget tables directly to each other.

Related: Uses modeling and conditional formatting.

Common Situations Interviewers Ask About

Definition: These are everyday problems you should be ready to explain.

Simple Explanation: Totals do not match the source. Report is slow. Refresh failed. Numbers differ between two reports. A user wants a change at the last minute.

Example: Report total is 5 lakh lower than Excel.

Technique (Step-by-Step): 1. Totals wrong: check filters, relationships, duplicates and data types. 2. Slow: Performance Analyzer. 3. Refresh failed: check credentials, gateway and source changes. 4. Different numbers: check if both reports use the same model and definition. 5. Last-minute change: ask what changed in the business need.

Interview Answer: “If totals do not match, I check filters first, then relationships and duplicates, then data types. I compare a small slice with the source until I find where the difference starts.”

Follow-up Qs:

- **Q:** What is your first step when a number looks wrong? **A:** Check it against the source for a small slice, like one day.
- **Q:** What do you do if you do not know the answer? **A:** Say so, explain how you would find out, and then do it.

Common Mistake: Saying 'I never make mistakes'. Everyone makes them. Explain how you find and fix them.

Related: Summary of the whole guide.

Level 11 - Quick Revision

Project	Main points
Sales	Sales, profit, YoY, region, top products
HR	Headcount, attrition, joiners, RLS
MIS	Automated daily report with refresh and email
Finance	Actual vs budget, variance, colors
Explain a project	Problem, data, what I did, result

Key rule: Say the problem, the data, what you did and the result.

Top Interview Questions: (1) Tell me about your project. (2) What was the hardest part? (3) What do you do when totals do not match?

Practical Scenarios: (1) Totals differ from Excel - check filters, relationships, duplicates. (2) Manager asks for a report in a day - reuse the model and keep it simple. (3) Interview: explain a dashboard - problem, data, work, result.

MASTER REVISION - ALL 11 LEVELS AT A GLANCE

Level	Topic	Core Items
1	Power BI Basics	What is Power BI, Desktop/Service/Mobile, licenses, views, workflow, file types
2	Getting Data	Data sources, Import, DirectQuery, Live Connection, how to choose
3	Power Query	Applied steps, data types, remove rows, split/merge, unpivot, group by, merge vs append, folder, errors
4	Data Modeling	Fact/dimension, star schema, relationships, cross-filter, date table, hierarchy, sort by column
5	DAX Basics	Column vs measure, context, CALCULATE, FILTER, ALL, DIVIDE, SWITCH, VAR, RELATED, SUMX
6	DAX Time and Ranking	YTD, last year, MoM, running total, % of total, RANKX, Top N
7	Visuals and Design	Chart choice, matrix, slicers, filters, conditional formatting, drill, bookmarks
8	Service and Sharing	Workspace, publish, dashboard, refresh, gateway, apps, RLS, subscriptions
9	Performance	Performance Analyzer, reduce size, measures, incremental refresh, naming
10	Analyst Skills	KPIs, requirements, wireframe, storytelling, variance, product metrics, MIS
11	Project Scenarios	Explain a project, sales, HR, MIS, finance, common problems

INTERVIEW IMPORTANT - DO NOT MISS THESE

- Desktop is for building. Service is for sharing.
- Import is the default. DirectQuery is for big or live data. Live Connection uses an existing model.
- Merge adds columns. Append adds rows.
- Reference stays linked to the original query. Duplicate is independent.
- Unpivot turns many columns into rows.
- Use a star schema: fact table in the middle, dimensions around it.
- Keep relationships one-to-many with Single direction.
- Time intelligence needs a proper Date table, marked as a Date table.
- Measures change with filters. Calculated columns are fixed row by row. Prefer measures.
- CALCULATE changes filters. ALL removes filters. DIVIDE avoids errors.
- Drill down goes deeper in the same visual. Drill through goes to another page.
- A report can have many pages. A dashboard is one page of pinned tiles.
- A gateway is needed for on-premises data refresh.
- RLS: create roles in Desktop, test with View as, add users in the Service.
- Slow report? Use Performance Analyzer first.
- Always check your totals against the source before sharing.

HOW TO USE THIS GUIDE

1. Go level by level in order. Later levels use ideas from earlier levels.

2. Download Power BI Desktop (it is free) and build a small sales report with a sample Excel file. Follow the steps in each topic by hand.

3. Read each Interview Answer, then say it out loud in your own words.

4. Try the Follow-up Questions before reading the answers.

5. Prepare one real project story using the Level 11 method: problem, data, what you did, result.
6. Before an interview, read the Quick Revision tables and the Interview Important list.

DAX FUNCTIONS - DEFINITIONS AND SCENARIOS

About This Section

Nobody uses all 250+ DAX functions. This section lists the functions that Data Analyst, Business Analyst and Power BI Developer roles are asked about. For each function you get a Definition you can say in an interview, and a Scenario that shows where it is used at work. When an interviewer asks "What does X do?", say the definition in one line, then give the scenario as your example.

Which Functions for Which Role

Role	Learn these parts	Main functions
Data Analyst	Part 1 first	SUM, AVERAGE, COUNTROWS, DISTINCTCOUNT, IF, SWITCH, IFERROR, CALCULATE, FILTER, ALL, VALUES, SELECTEDVALUE, DIVIDE, text and date functions
Business Analyst	Part 1, then Part 2	TOTALYTD, SAMEPERIODLASTYEAR, DATEADD, DATESINPERIOD, PARALLELPERIOD, RANKX, TOPN, ALLSELECTED, MAXX, MINX
Power BI Developer	Parts 1, 2 and 3	ALLEXCEPT, REMOVEFILTERS, KEEPFILTERS, SUMX, SUMMARIZE, ADDCOLUMNS, TREATAS, USERELATIONSHIP, LOOKUPVALUE, VAR and RETURN, ISINSCOPE, USERPRINCIPALNAME, calculation groups, WINDOW, OFFSET, INDEX

Learning order: CALCULATE and filter context, then time intelligence, then iterators (SUMX), then VAR and RETURN, then table functions, then window functions.
Common Mistake: Memorizing the definition without knowing a scenario. Interviewers always follow up with "Where have you used it?"

PART 1 - DATA ANALYST CORE FUNCTIONS

Aggregation Functions

SUM

Definition: Adds up all the numbers in one column.
Scenario: A manager wants Total Sales in a card at the top of the report: Total Sales = SUM(Sales[Amount]).

AVERAGE

Definition: Gives the mean (average) of the numbers in a column.
Scenario: Find the average order value for each region to see which region has bigger baskets.

MIN

Definition: Gives the smallest value in a column.
Scenario: Find the first order date of a customer, or the lowest price a product was sold at.

MAX

Definition: Gives the largest value in a column.
Scenario: Find the latest order date in the data, or the highest single sale in a month.

MEDIAN

Definition: Gives the middle value of a column, so a few very large values do not pull the answer up.
Scenario: Show the typical delivery time when a few orders took 30 days and would distort the average.

COUNT

Definition: Counts the cells in a column that hold numbers or dates. It skips blanks and text.
Scenario: Count how many orders have an amount filled in.

COUNTA

Definition: Counts all the non-blank values in a column, including text.
Scenario: Count how many customers have an email address entered in the customer table.

COUNTROWS
Definition: Counts the rows in a table.
Scenario: Number of orders: Orders = COUNTROWS(Sales). It also works on a filtered table.

DISTINCTCOUNT
Definition: Counts the unique values in a column, ignoring repeats.
Scenario: Count unique customers who bought this month, even if one customer placed many orders.

Logical and Error Handling Functions

IF
Definition: Checks one condition and returns one value if it is true and another if it is false.
Scenario: Mark each order as Big if the amount is over 1000, otherwise Small.

SWITCH
Definition: Checks many conditions in order and returns the result of the first one that is true. SWITCH(TRUE(), ...) replaces long nested IFs.
Scenario: Label every region as High, Medium or Low based on its sales, or turn marks into grades.

AND
Definition: Returns true only when both conditions are true.
Scenario: Select orders where Region is West and Amount is above 1000.

OR
Definition: Returns true when at least one of the conditions is true.
Scenario: Mark customers from Delhi or Mumbai as Metro customers.

NOT
Definition: Reverses a condition: true becomes false and false becomes true.
Scenario: Leave out cancelled orders from a count: NOT(Sales[Status] = "Cancelled").

IFERROR
Definition: Returns a value you choose when a calculation gives an error.
Scenario: Show 0 instead of an error when a text value cannot be turned into a number.

ISBLANK
Definition: Checks if a value is blank and returns true or false.
Scenario: Show 'Not Shipped' in the report when the Ship Date is blank.

BLANK
Definition: Returns an empty value.
Scenario: Return nothing, instead of 0, when there is no data, so the chart does not show useless zero bars.

COALESCE
Definition: Returns the first value in a list that is not blank.
Scenario: Use actual sales if they exist, otherwise use the forecast: COALESCE([Actual], [Forecast]).

IN
Definition: Checks whether a value is inside a list of values.
Scenario: Filter two regions in one condition: Region[Name] IN {"West", "East"}.

Filter Basics

CALCULATE
Definition: Changes the filters and then calculates a value. It is the most important DAX function.
Scenario: Show West region sales whatever the slicer says, or show last year's sales next to this year's.

FILTER
Definition: Returns only the rows of a table that meet a condition.
Scenario: Total sales of customers who spent more than 10,000 in total. The condition needs a row-by-row check, so FILTER is needed.

ALL
Definition: Removes filters from a table or a column.
Scenario: Percentage of total: divide each product's sales by the sales of all products together.

VALUES
Definition: Returns the unique values that are left in a column after the current filters.
Scenario: Count how many regions are currently selected: COUNTROWS(VALUES(Region[Name])).

SELECTEDVALUE
Definition: Returns the value if only one value is picked, otherwise it returns a default value you give.
Scenario: Make a chart title that changes with the slicer: 'Sales for West', or 'All Regions' when nothing is picked.

Math Functions

DIVIDE
Definition: Divides one number by another and safely handles division by zero by giving blank or a value you choose.
Scenario: Profit Margin % = DIVIDE([Profit], [Sales]) without errors for products that have zero sales.

ROUND
Definition: Rounds a number to the number of decimal places you choose.
Scenario: Show average price rounded to 2 decimals in a table.

ABS
Definition: Gives the number without its minus sign.
Scenario: Show the size of the gap between actual and target, whether the team is over or under.

INT
Definition: Rounds a number down to the nearest whole number.
Scenario: Work out completed years of service from days: INT(Days / 365).

Text Functions

LEFT
Definition: Takes the first N characters from a text.
Scenario: Take the first 3 letters of a product code to get the category code.

RIGHT
Definition: Takes the last N characters from a text.
Scenario: Take the last 4 digits of an ID number.

MID
Definition: Takes characters from the middle of a text, starting at a position you give.
Scenario: Pull the region code out of an order number like ORD-WST-1001.

LEN
Definition: Counts the number of characters in a text.
Scenario: Check that every phone number has exactly 10 digits and flag the ones that do not.

UPPER and LOWER
Definition: UPPER changes text to capital letters. LOWER changes it to small letters.
Scenario: Make city names the same before comparing them, so 'delhi' and 'DELHI' match.

TRIM
Definition: Removes extra spaces from a text.
Scenario: Clean customer names with hidden spaces so one customer does not appear twice in a chart.

CONCATENATE
Definition: Joins two pieces of text into one. The & sign does the same and can join more than two.
Scenario: Make a full name from first name and last name: [First] & " " & [Last].

SEARCH
Definition: Finds the position of one text inside another. It is not case-sensitive.
Scenario: Find the position of '@' in an email address so you can cut out the domain name.

SUBSTITUTE
Definition: Replaces a piece of text with another piece of text.
Scenario: Replace '-' with '/' in product codes so they match the other system.

FORMAT
Definition: Converts a value into text in the format you choose. The result is text, not a number.
Scenario: Show a date as the short month name: FORMAT('Date'[Date], "MMM").

CONTAINSSTRING
Definition: Returns true if a text contains another piece of text. It is not case-sensitive.
Scenario: Flag every product whose name contains the word 'Pro'.

Date Functions

DATE
Definition: Builds a date from a year, a month and a day.
Scenario: Set the first day of a date table: DATE(2022, 1, 1).

YEAR, MONTH and DAY
Definition: Give the year, the month number or the day number of a date.
Scenario: Create the Year and Month No columns in a Date table.

WEEKDAY
Definition: Gives the day of the week as a number. By default Sunday is 1.
Scenario: Compare weekday sales with weekend sales.

TODAY
Definition: Gives today's date. It updates when the report is opened or refreshed.
Scenario: Days since an order was placed: DATEDIFF(Sales[OrderDate], TODAY(), DAY).

NOW
Definition: Gives the current date and time. In the Power BI Service it shows UTC time.
Scenario: Show a 'Last refreshed' time on the report.

DATEDIFF
Definition: Gives the number of days, months or years between two dates.
Scenario: Delivery days = DATEDIFF(Sales[OrderDate], Sales[DeliveryDate], DAY), or employee tenure in months.

EOMONTH
Definition: Gives the last date of a month, moved forward or back by the months you give.
Scenario: Find the month-end date for month-end reports or for invoice due dates.

PART 2 - BUSINESS ANALYST DAX FUNCTIONS

Time Intelligence Functions

TOTALYTD
Definition: Gives the total from the start of the year up to the current date. It needs a proper Date table.
Scenario: The sales head wants sales so far this year, shown month by month.

TOTALQTD and TOTALMTD
Definition: Give the total from the start of the quarter (QTD) or the start of the month (MTD) up to the current date.
Scenario: Quarter-to-date sales for the quarterly review and month-to-date sales for daily target tracking.

DATESYTD
Definition: Returns the list of dates from the start of the year to the current date. It is used inside CALCULATE.
Scenario: Year-to-date sales for West region only: CALCULATE([Total Sales], DATESYTD('Date'[Date]), Region[Name] = "West").

SAMEPERIODLASTYEAR
Definition: Returns the same dates one year earlier.
Scenario: The CEO wants March this year compared with March last year, to calculate YoY growth.

DATEADD
Definition: Moves a set of dates forward or back by days, months, quarters or years.
Scenario: Previous month sales for the MoM comparison: DATEADD('Date'[Date], -1, MONTH).

PARALLELPERIOD
Definition: Returns the full month, quarter or year that is shifted forward or back from the current dates, even when only part of the period is selected.
Scenario: Compare this quarter with the whole previous quarter.

PREVIOUSMONTH
Definition: Returns all the dates of the month before the current one.
Scenario: Show last month's sales on a KPI card next to this month.

DATESINPERIOD
Definition: Returns the dates from a start date going a number of days, months, quarters or years forward or back.
Scenario: Rolling 3-month sales that end at the current month.

DATESBETWEEN
Definition: Returns the dates between a start date and an end date that you give.
Scenario: Sales between the start and the end date of a festival offer.

Ranking, Top N and Comparison Functions

RANKX
Definition: Gives a rank number (1, 2, 3...) to each item based on a value.
Scenario: Rank every store by sales, with rank 1 for the best store.

TOPN
Definition: Returns the top N rows of a table based on a value.
Scenario: Show the total sales of the top 5 customers, or a top 10 products table.

MAXX and MINX
Definition: Calculate an expression for each row of a table and return the highest (MAXX) or lowest (MINX) result.
Scenario: Find the sales of the best month: MAXX(VALUES('Date'[Month]), [Total Sales]). Use MINX for the worst month.

ALLSELECTED
Definition: Removes the filters of the visual but keeps the choices the user made in slicers.
Scenario: Percentage of total that stays inside the years the user selected. It is also used in running totals.

PART 3 - POWER BI DEVELOPER DAX FUNCTIONS

Filter Control Functions

ALLEXCEPT
Definition: Removes all filters from a table except the columns you name.
Scenario: Sales share of a product inside its own category: remove all filters but keep the Category filter.

REMOVEFILTERS
Definition: Clears filters inside CALCULATE. It does the same job as ALL but the name makes the intention clearer.
Scenario: Grand total in a % of total measure: CALCULATE([Total Sales], REMOVEFILTERS(Product)).

KEEPFILTERS
Definition: Adds a new filter without replacing the filter the user already has on the same column.
Scenario: Show West sales only if West is inside the user's slicer selection. If the user picked East, the result is blank.

CROSSFILTER
Definition: Changes the cross-filter direction of a relationship for one measure only.
Scenario: Count customers by the product they bought using Both direction inside one measure, instead of changing the whole model to Both.

Iterator Functions (X Functions)

SUMX
Definition: Goes row by row, calculates an expression for each row, and then adds all the results.
Scenario: Revenue = SUMX(Sales, Sales[Qty] * Sales[Price]). SUM(Qty) * SUM(Price) would give a wrong answer.

AVERAGEX
Definition: Calculates an expression for each row and then gives the average of the results.
Scenario: Average sales per customer: AVERAGEX(VALUES(Customer[Name]), [Total Sales]).

COUNTX
Definition: Counts the rows where the expression is not blank.
Scenario: Count the customers who actually have sales: COUNTX(Customer, [Total Sales]).

Table Functions

SUMMARIZE
Definition: Creates a summary table that is grouped by the columns you choose.
Scenario: Build a table of sales per customer and product to use inside another calculation.

SUMMARIZECOLUMNS
Definition: Groups data and adds calculated values in one step. It is the fast, modern way, and it is what Power BI visuals run behind the scenes.
Scenario: Create a calculated table with sales by region and month.

ADDCOLUMNS
Definition: Adds new calculated columns to a table in memory.
Scenario: Take a list of customers and add Total Sales and Rank beside each one before filtering.

SELECTCOLUMNS
Definition: Returns a table with only the columns you choose, and lets you rename them.
Scenario: Make a small lookup table with only CustomerID and Name.

GROUPBY
Definition: Groups a table and calculates values for each group with an iterator such as SUMX over CURRENTGROUP().
Scenario: Group a table that you built in DAX by category. For normal tables SUMMARIZE is usually easier.

DISTINCT
Definition: Returns the unique values of a column, or the unique rows of a table.
Scenario: Make a list of unique cities: DISTINCT(Customer[City]).

UNION
Definition: Puts two tables on top of each other. Both need the same number of columns.
Scenario: Combine an Actual table and a Budget table into one table for a single chart.

INTERSECT
Definition: Returns the rows that appear in both tables.
Scenario: Find customers who bought in both 2024 and 2025 (repeat customers).

EXCEPT
Definition: Returns the rows of the first table that are not in the second table.
Scenario: Find lost customers: those who bought last year but not this year.

CROSSJOIN
Definition: Pairs every row of one table with every row of another table.
Scenario: Build all Product and Month combinations, so products with zero sales still show in a report.

GENERATESERIES
Definition: Creates a table of numbers from a start value to an end value with a step.
Scenario: Make a discount slider with values 0 to 50 in steps of 5 for a what-if analysis.

ROW
Definition: Creates a table with one row and named values.
Scenario: Return several KPI values in one row to test them quickly in DAX query view.

CONCATENATEX
Definition: Joins the text of many rows into one text, with a separator.
Scenario: Show the selected regions as 'West, East' in a title.

Relationship Functions

RELATED
Definition: Brings a value from the 'one' side table into the current row of the 'many' side table. It needs a relationship.
Scenario: Pull Category from the Product table into the Sales table.

RELATEDTABLE
Definition: Returns the related rows from the 'many' side for the current row.
Scenario: Number of orders for each customer in a calculated column: COUNTROWS(RELATEDTABLE(Sales)).

USERELATIONSHIP
Definition: Uses an inactive relationship for one calculation.
Scenario: Sales by Ship Date while the report is normally linked on Order Date.

LOOKUPVALUE
Definition: Finds a value in a table by matching columns. It works without a relationship.
Scenario: Get the price from a price list table that is not linked to the Sales table.

TREATAS
Definition: Uses the values of one table as a filter on columns of another table, without a physical relationship.
Scenario: Apply the department codes of a Budget table as filters on the Sales table.

Variables and Filter Context Functions

VAR and RETURN
Definition: VAR stores a value under a name inside a measure. RETURN gives the final result. The value is calculated once.
Scenario: Growth % with VAR Current and VAR Previous, so the formula is short and easy to read.

HASONEVALUE
Definition: Returns true when only one value is left in a column under the current filters.
Scenario: Show customer details only when exactly one customer is selected. Otherwise show blank.

ISFILTERED
Definition: Returns true when a column is directly filtered.
Scenario: Show one result when the user has picked a region in the slicer and a different one when nothing is picked.

ISINSCOPE
Definition: Returns true when a column is the level that is being shown in a matrix row or column.
Scenario: Show a different value at Year level and at Month level in a matrix, or fix totals and subtotals.

EARLIER
Definition: Reaches the outer row in a nested row context in a calculated column. VAR is now preferred.
Scenario: The old way to rank a row inside its own category in a calculated column. In an interview, say you would use VAR.

Security, Calculation Groups and Date Table Functions

USERNAME
Definition: Returns the login name of the current user, in the form DOMAIN\user.
Scenario: Works for RLS in Desktop. In the Service, USERPRINCIPALNAME is the safer choice.

USERPRINCIPALNAME
Definition: Returns the email (UPN) of the person who is viewing the report.
Scenario: Dynamic RLS: filter a Security table where Email = USERPRINCIPALNAME(), so each manager sees only their own region.

SELECTEDMEASURE
Definition: Used in calculation groups. It returns the measure that is currently being calculated.
Scenario: One calculation item for YTD works for every measure: TOTALYTD(SELECTEDMEASURE(), 'Date'[Date]).

SELECTEDMEASURENAME
Definition: Returns the name of the measure being calculated, as text.
Scenario: Choose a different number format for each measure inside a calculation group, such as percent or currency.

CALENDAR
Definition: Creates a Date table with one row for each date between a start date and an end date.
Scenario: Build a Date table from 2022 to 2025: CALENDAR(DATE(2022,1,1), DATE(2025,12,31)).

CALENDARAUTO
Definition: Creates a Date table automatically from the dates found in the model, in whole years.
Scenario: A quick Date table when the date range of the data changes with every refresh.

Window Functions (Newer)

WINDOW
Definition: Returns the rows that fall inside a range of positions in an ordered table.
Scenario: Moving average of the last 3 months.

OFFSET
Definition: Returns the row that is a given number of positions before or after the current row.
Scenario: Previous row comparison, such as last month's value next to this month, without time intelligence functions.

INDEX
Definition: Returns the row at a fixed position in an ordered table, such as first or last.
Scenario: Show the sales of the first month or the last month in the list, or the best day of the month.

DAX Functions - Quick Revision

Key rule: CALCULATE changes filters, ALL and REMOVEFILTERS clear them, DIVIDE avoids errors, and time functions need a proper Date table.
Top Interview Questions: (1) What does CALCULATE do? (2) SUM vs SUMX. (3) ALL vs ALLEXCEPT vs ALLSELECTED. (4) How do you calculate YoY growth? (5) What does USERELATIONSHIP do? (6) How do you write dynamic RLS?

TOP 100 POWER BI INTERVIEW QUESTIONS WITH ANSWERS

Read each answer, then say it in your own words. Short code is shown where an interviewer may ask for a formula.

A. POWER BI BASICS

Q1. What is Power BI?

A: Power BI is a Microsoft tool that helps me take data from different places, clean it and show it as charts and dashboards. It helps managers understand numbers quickly.

Q2. Why do companies use Power BI?

A: It is easy to use, connects to many data sources, makes interactive reports and is easy to share. It also costs less than many other BI tools.

Q3. What are the main parts of Power BI?

A: Power BI Desktop to build reports, Power BI Service (online) to publish and share, and the Mobile app to view reports. There is also a Gateway for company data.

Q4. What is the difference between Power BI Desktop and Power BI Service?

A: I build reports in Desktop. I publish them to the Service so others can view them, set refresh and share.

Q5. Is Power BI Desktop free?

A: Yes, Desktop is free. To share reports in workspaces, people usually need a Pro license.

Q6. What is the difference between Excel and Power BI?

A: Excel is great for calculations and small data. Power BI is made for bigger data, interactive reports and easy sharing.

Q7. What are the three views in Power BI Desktop?

A: Report view to build charts, Table (Data) view to check the data, and Model view to connect tables.

Q8. What is a .pbix file?

A: It is the Power BI file that holds the report, the data and the model.

Q9. What is a semantic model?

A: It is the data and model behind a report. It used to be called a dataset. Many reports can use one model.

Q10. What is the Power BI workflow you follow?

A: I understand the need, get the data, clean it in Power Query, build the model, write measures, create visuals, publish, and set refresh.

Q11. What is the difference between a report and a dashboard?

A: A report can have many pages and comes from one model. A dashboard is one page of pinned tiles from one or many reports, made in the Service.

Q12. What is a Power BI workspace?

A: It is a shared area in the Service where a team keeps its reports, dashboards and models.

B. GETTING DATA AND CONNECTIVITY

Q13. What data sources have you used?

A: Mostly Excel, CSV, SQL Server and SharePoint folders. I have also used web data.

Q14. What is Import mode?

A: Power BI copies the data into the model. It is fast, but the data is only as fresh as the last refresh.

Q15. What is DirectQuery?

A: Power BI does not store the data. It asks the database every time. It is good for big or live data, but it can be slower.

Q16. What is Live Connection?

A: I connect to a model that is already published, so I only build visuals and cannot change the model.

Q17. Import vs DirectQuery: which would you choose?

A: I choose Import by default because it is fast. I use DirectQuery only if the data is too big or must be live.

Q18. What is a Composite model?

A: A model that uses both Import and DirectQuery tables together.

Q19. What is a gateway?

A: A gateway is a bridge that lets the Service refresh data from files or databases inside the company network.

Q20. Do you need a gateway for SharePoint Online or Azure SQL?

A: Usually not, because they are in the cloud. On-premises SQL Server or local files do need a gateway.

C. POWER QUERY

Q21. What is Power Query?

A: It is the tool inside Power BI where I clean and shape the data before loading it.

Q22. What are Applied Steps?

A: They are the saved list of every change I make. Every time data refreshes, the same steps run again.

Q23. Merge vs Append?

A: Merge joins two tables side by side using a matching column, like a lookup. Append puts two tables one below the other, like monthly files.

Q24. Reference vs Duplicate?

A: Reference makes a copy that is linked to the original query. Duplicate makes an independent copy of all the steps.

Q25. What is Unpivot and when do you use it?

A: Unpivot turns columns into rows. I use it when each month is a separate column, so I get one Month column and one Value column.

Q26. What is Group By?

A: It summarizes rows into groups, like total sales per customer, before loading.

Q27. How do you remove duplicates?

A: Select the column that should be unique, like Order ID, and use Home, Remove Rows, Remove Duplicates.

Q28. How do you handle null values?

A: I check what null means for the business. Then I replace it with 0 or Unknown, filter it out, or fill it from other rows.

Q29. How do you fix wrong data types?

A: I click the icon on the column and choose the right type, such as Date or Decimal number. I do it early in the steps.

Q30. What is a Conditional Column?

A: A click-based way to add a column using if-then rules, like High if sales is above 1000, else Low.

Q31. How do you combine many files with the same layout?

A: I use Get data from Folder, then Combine and Transform. New files added to the folder are picked up on refresh.

Q32. Which join kinds are in Merge Queries?

A: Left Outer, Right Outer, Full Outer, Inner, Left Anti and Right Anti. I use Left Outer most.

Q33. What is query folding?

A: It means Power Query sends your steps back to the database as one SQL query, so the database does the work. It makes refresh faster.

Q34. What is M language?

A: It is the formula language behind Power Query. Most of the time I use the buttons and Power BI writes the M for me.

D. DATA MODELING

Q35. What is a star schema?

A: A model with one fact table in the middle and dimension tables around it, like a star. It is simple and fast.

Q36. What is a fact table and a dimension table?

A: A fact table holds numbers like sales. A dimension table describes things like product, customer and date.

Q37. What is cardinality?

A: It says how rows in two tables match: one-to-many, many-to-one, one-to-one or many-to-many.

Q38. Which relationship type is most common?

A: One-to-many, from a dimension to a fact table.

Q39. What is cross-filter direction?

A: It says which way filters flow between tables. Single is normal. Both lets filters flow both ways but can cause problems.

Q40. Why do you avoid Both direction?

A: It can slow the report and give unexpected numbers.

Q41. Active vs inactive relationship?

A: Only one relationship between two tables can be active. Inactive ones are dotted, and I use USERELATIONSHIP in a measure to use them.

Q42. Why do you need a date table?

A: Time functions like year to date and last year need a proper table with one row for every date.

Q43. How do you create a date table?

A: Use CALENDAR or CALENDARAUTO in a new table, add Year and Month columns, link it to the fact table and mark it as a date table.

```
Date = CALENDAR(DATE(2022,1,1), DATE(2025,12,31))
```

Q44. What is a hierarchy?

A: A path of fields like Year, Quarter, Month, so users can drill down.

Q45. Why do months show in the wrong order?

A: Month names sort alphabetically. I use Sort by Column and sort Month Name by Month Number.

Q46. What is a role-playing dimension?

A: One dimension used for different purposes, like one Date table used for both Order Date and Ship Date, with one active and one inactive relationship.

E. DAX

Q47. What is DAX?

A: DAX is the formula language of Power BI. I use it to write measures, calculated columns and calculated tables.

Q48. Calculated column vs measure?

A: A column is calculated row by row and stored. A measure is calculated when used and changes with filters. I use measures most of the time.

Q49. What is a calculated table?

A: A new table I create using a DAX formula, like a date table.

Q50. What is row context?

A: It means DAX works on one row at a time, like in a calculated column.

Q51. What is filter context?

A: It is the set of filters at the moment of calculation, from slicers, filters and chart axes. It is why one measure gives different values in a chart.

Q52. What does CALCULATE do?

A: It changes the filters and then calculates a value.

```
West Sales =
CALCULATE([Total Sales], Region[Name] = "West")
```

Q53. What is the difference between SUM and SUMX?

A: SUM adds one column. SUMX goes row by row, calculates something like Qty times Price, and then adds the results.

Q54. COUNT vs COUNTROWS vs DISTINCTCOUNT?

A: COUNT counts values in a column. COUNTROWS counts rows in a table. DISTINCTCOUNT counts unique values.

Q55. What does ALL do?

A: It removes filters from a table or column. I use it for totals and percentages of total.

Q56. ALL vs ALLEXCEPT vs ALLSELECTED?

A: ALL removes every filter. ALLEXCEPT removes all filters except the columns I name. ALLSELECTED removes filters but keeps what the user chose in slicers.

Q57. What does FILTER do?

A: It returns only the rows that meet a condition. I use it inside CALCULATE when the condition is more than a simple column check.

Q58. Why use DIVIDE instead of the / sign?

A: DIVIDE handles division by zero without errors.

Q59. What are VAR and RETURN?

A: VAR stores a value in a measure and RETURN gives the final result. It makes formulas easier to read.

```
Growth % =
VAR Curr = [Total Sales]
VAR Prev = [Sales LY]
RETURN DIVIDE(Curr - Prev, Prev)
```

Q60. How do you calculate year to date?

A: I use TOTALYTD with the date column from the Date table.

```
Sales YTD =
TOTALYTD([Total Sales], 'Date'[Date])
```

Q61. How do you calculate last year sales and YoY growth?

A: I use SAMEPERIODLASTYEAR inside CALCULATE for last year, then DIVIDE for the percent change.

```
Sales LY =
CALCULATE([Total Sales],
  SAMEPERIODLASTYEAR('Date'[Date]))

YoY % = DIVIDE([Total Sales] - [Sales LY], [Sales LY])
```

Q62. How do you calculate month over month?

A: I use DATEADD with -1 and MONTH to get last month, then compare.

```
Sales PM =
CALCULATE([Total Sales],
  DATEADD('Date'[Date], -1, MONTH))
```

Q63. How do you calculate percentage of total?

A: Divide the item sales by total sales calculated with ALL.

```
% of Total =
DIVIDE([Total Sales],
  CALCULATE([Total Sales], ALL(Product)))
```

Q64. How do you rank products by sales?

A: I use RANKX over all products with the sales measure.

```
Rank =
RANKX(ALL(Product[ProductName]), [Total Sales], , DESC)
```

Q65. What does RELATED do?

A: It brings a value from the related table, like a lookup. It needs a relationship.

Q66. What does SELECTEDVALUE do?

A: It returns the value if only one is selected in a slicer. I use it for dynamic titles.

F. VISUALS AND REPORT DESIGN

Q67. How do you choose the right chart?

A: I think about the question. Bars to compare, lines for time, cards for one number, tables for details and maps for places.

Q68. Table vs Matrix?

A: A table is a plain list. A matrix works like a pivot table with rows, columns and totals.

Q69. What is a slicer?

A: A filter that sits on the page and lets users click to choose values.

Q70. Slicer vs filter?

A: A slicer is visible and the viewer uses it. A filter in the Filters pane is set by the report builder.

Q71. What are the filter levels in Power BI?

A: Visual level, page level and report level.

Q72. What is drill down?

A: Going deeper inside the same chart, like Year to Quarter to Month.

Q73. What is drill through?

A: Jumping to another page with details about the item I selected, already filtered.

Q74. What are bookmarks?

A: They save a page state, like filters and visible visuals. I use them with buttons for toggles and resets.

Q75. What is conditional formatting?

A: Changing colors or adding icons based on values, like red for below target.

Q76. What is a tooltip?

A: A small box that shows extra details when I hover over a visual.

Q77. What are good design tips for a dashboard?

A: Key numbers on top, few colors, clear titles, not too many visuals and consistent alignment.

Q78. What is Edit interactions?

A: It lets me choose which visuals are affected when I click on another visual or slicer.

G. POWER BI SERVICE AND SECURITY

Q79. How do you publish a report?

A: In Desktop, click Publish on the Home tab and choose the workspace.

Q80. What is scheduled refresh?

A: It updates the data in the Service automatically at set times. Pro allows up to 8 per day.

Q81. What do you do if the refresh fails?

A: I check the error message, credentials, gateway status and whether the source file or table changed.

Q82. What is row-level security?

A: It makes sure each person only sees the data they are allowed to see.

```
[Region] = "West"
```

Q83. How do you set up RLS?

A: In Desktop, Modeling, Manage roles, add a filter, test with View as, publish, then add users to the role in the Service.

Q84. Static vs dynamic RLS?

A: Static uses a fixed filter per role. Dynamic uses the user's login with USERPRINCIPALNAME, so one role works for everyone.

Q85. What is a Power BI app?

A: A packaged set of reports and dashboards for a large audience, read-only and neat.

Q86. What are workspace roles?

A: Admin, Member, Contributor and Viewer. Each can do different things, and Viewer can only view.

Q87. Do viewers need a Pro license?

A: Usually yes, unless the content is in a Premium or Fabric capacity workspace.

Q88. What are subscriptions and alerts?

A: A subscription emails a report on a schedule. An alert tells me when a dashboard number crosses a limit.

Q89. What is Analyze in Excel?

A: It lets me connect Excel to a published model and use pivot tables on it.

Q90. Can you export data from Power BI?

A: Yes, to Excel, CSV, PDF or PowerPoint, if the report settings allow it.

H. PERFORMANCE AND BEST PRACTICE

Q91. How do you make a slow report faster?

A: I use Performance Analyzer to find the slowest visual, remove unused columns and rows, use fewer visuals per page, prefer measures and keep a simple star schema.

Q92. What is Performance Analyzer?

A: A tool in the View tab that shows how long each visual takes to load.

Q93. What is incremental refresh?

A: It refreshes only the new or recent data instead of the whole table. It is good for big tables.

Q94. Why avoid calculated columns when possible?

A: They are stored and use memory. Measures are calculated only when needed.

Q95. How do you check that your numbers are correct?

A: I compare totals with the source, check a small slice by hand and ask a colleague to review.

I. BUSINESS AND SCENARIO QUESTIONS

Q96. How do you gather requirements for a dashboard?

A: I talk to the user, ask what decisions they take, list the KPIs and filters, check the data source and confirm it in writing.

Q97. What is a KPI?

A: A key number that shows progress toward a goal, like monthly sales against target.

Q98. Tell me about a Power BI project you did.

A: I explain the business problem, the data, what I did in Power Query, model and DAX, the visuals and the result. I use my own real project here.

Q99. Totals in your report do not match the source. What do you do?

A: I check filters, relationships, duplicates and data types. I compare a small slice with the source to find where the difference starts.

Q100. A user asks for a change at the last minute. What do you do?

A: I ask what the business needs, check how big the change is, tell them how long it will take, and do it carefully without breaking other things.